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BEFORE YOU BEGIN THE SOFTWARE LOAD - CAUTIONS AND WARNINGS

Important: If your system is connected to a Hospital Network, disconnect it before proceeding. Failure to disconnect the SGI computer can corrupt the software load process.

Important: Disconnect the DASM from the host or remove power from the DASM before beginning the load and remember to reconnect it after the load has finished. The OnLine Center Support has reported intermittent system lockups on Octane computers during software loads with the DASM connected.

Important: During the Install GUI phase of the software load, be sure the **<Num Lock>** key on the key board is turned off. If the **<Num Lock>** is on, the symptom seen is that nothing will happen when trying to type in any of the edit fields in the Install GUI.

Important: When reconfiguring the system using the Install/Configuration GUI, remember to push the **<Configure>** button on the verify tab before going to the Save info tab and doing a SaveInfo. If you have not done the final configuration by pushing the **<Configure>** button in the Verification tab, the information saved during Save Info will not reflect the latest changes made in the Install GUI.

Important: If at any time during the OS load a problem occurs that causes you to stop the install, correct the problem then reset the computer or power cycle the computer and begin the OS install again from the start. This is necessary to insure a complete load. If this does not fix your problem it may be necessary to do a low level format on the system drives. Refer to Appendix C for this process.

Important: If the site will be using the "YP" networking option, **YP must be turned off**, until after the InSite IIP option has been configured and checked out, or it will cause the InSite checkout to fail.

Important: If at the end of OS load the system will not boot and stops with the error, "/var/root partition corruption". Do the following:

1. Open a shell window where by typing **sh <Enter>**.
2. Type at the prompt: **dd if=/dev/zero of=/dev/rdisk/dks0d1s0 count=1000 <Enter>**.
This writes zeros across the disk and will run for less than one minute.
3. Start the OS load from the beginning.

This error is a very rare event.

Important: Release ASP, 9.0 and 9.1 can only be loaded on Octane Computer Systems.

Important: If you are reloading a system that is using two 9 Gigabyte disk drives, you must shutdown the Octane computer and remove the middle 9 Gigabyte drive before continuing the software load. After the load from cold, you will shutdown and reinstall the middle 9 Gigabyte and bring the system backup, start Guided Install and configure the drive back into the system. Steps have been added in this document to remind the person loading the software when to remove and reinstall the second 9 Gigabyte disk drive.

1- LOADING HOST COMPUTER - OVERVIEW & PREPARATION

Note

If this is a new install or system hardware upgrade of a Signa system, the system software has been pre-loaded and will not need to be reloaded. Skip to section 4 for information on how to reconfigure the system for new installs or upgrades.

1-1 Required Media

- IRIX 6.5 Operating System CDROM Set, part number 2352720
- MrApps - MR Applications CDROM, part number 2362853
- MrpResSrv - MR Restricted Service CDROM (*GE Service use only*)
or MrpAdvSrv - MR Advanced Service CDROM (*Licensed In-House Customers*)
- saveINFO - MaxOptix MOD Media (Either blank or previously used)

DO NOT PROCEED UNLESS YOU HAVE ALL THIS MEDIA AVAILABLE

1-2 Archive Images

This installation instruction will erase any current images left in the database. **Archive all images BEFORE continuing with this procedure.**

1-3 Required Information Prior To Loading Software:

Vital site information is recorded to ensure proper software installation and configuration of the new software.

1. With the system up and running, open The Common Service Desktop (CSD). Select **Configuration** tab, then select **Guided Install / FE Mode**.
2. Then select **Click here to start this tool**. The password is **operator**.
3. Step through each tab. Record the values from each field using the Datasheet (Table 1-1).
4. The following parameters are determined by physical observation. Record results in Table 2-1.
 - **Package Name** (Echospeed SR150, Echospeed SR120, Highspeed SR77, Base SR20)
 - **Modem Type**
 - **Camera Type**
 - **DASM Type**
 - **Coil Type** (BRM or CRM)
5. If this is a Mobile System, select **[Mobile]**, then record all **Site Names** and **IP addresses**.

TABLE 1-1
SITE CONFIGURATION INFORMATION

TAB	ENTRY (✓- Vital Info)	VALUE
Network Info	✓ (Top of Network tab)	__ Network Computer __ Standalone Computer
	✓ Host IP Address	
	✓ Netmask	(Default is 255.255.252.0)
	✓ Hostname	
	✓ Domain Name	
	YP(NIS) Service	__ On __ Off
	✓ IWS PC IP Address	
	Magnet Monitor IP Address	_____ (This parameter not used for anything but OpenSpeed systems. However, you must put in a placeholder value: (e.g. 0.0.0.0))
	✓ FE Laptop IP Address	
	✓ TPS Subnet (IP)	
	✓ Default Router IP Address	
	Power Monitor PC IP Address	_____ (This parameter not used for anything but 3T systems. However, you must put in a placeholder value: (e.g. 0.0.0.0))
System Configure	✓ Hospital Name	
	✓ Suite ID	
	✓ Host ID	
	✓ Unique ID	
	✓ Service ID	
	Date Format	
	Weight Unit	__ Pounds __ Kilograms
	Monitor Type	__ Gray __ Color
	Palette	__ Gray __ Color
	Keyboard	__ Amer. __ Eng. __ Fr. __ Ger. __ Ital. __ Port. __ Span. __ Swed.
	Language	__ Eng. __ Fr. __ Ger. __ Ital. __ Port. __ Span.
	Display Type	CRT/Custom value ie. 1.1
Hardware Configure	✓ Field Strength	__ 0.35T __ 0.5T __ 0.7T __ 1.0T __ 1.5T __ 3.0T __ 4.0T
	✓ Package Name	__ ECHOSPEED SR120 __ ECHOSPEED SR150 __ HIGHSPEED SR77 __ Base (Horizon)SR20 __ SMARTSPEED SR50
	✓ Gradient Type	__ 8645 (Single or Double Bay) __ 8280 (SGD Base System) __ 8651 (SGD HighSlew) __ 8915 (ACGD)

TAB	ENTRY (✓ - Vital Info)	VALUE
Hardware Configuration (Continued)	✓ Resonance Module	BRM CRM
	Magnet Serial Number	
	Table Limit	
	Line Frequency	60Hz 50Hz
	✓ RF Amp Type	Tube Type RF PEN II 1.0T Teli Solid State RF SRFD 1.5T Analogic Solid State RF SRFD
	✓ Governing Body	iec mhw1 mhw2 fda2
	✓ CERD (KHz) (See Note 1 at the end of this table for explanation of values)	CERD 1/4 Ch 62.5 62.5 62.5 62.5 CERD1/4 Ch 125 125 125 125 CERD 4 Ch 125 62.5 62.5 62.5 CERD 4 Ch 125 125 62.5 62.5 CERD 4 Ch 125 125 125 62.5 UCERD 1/4 Ch 125 125 125 125
	✓ ISO Vector Z (also record actual site value)	8981 12520 15250 (actual site value)
	MAGNET ENCLOSURE	Pre wide open(box style) or wide open (tight fit)
	PAC TYPE	Pac 1 or Pac 2
Edit Patient Data	Edit Patient Data Time Limit	
	Max Images Installed Per Update	
	Log File Header	Log File Header (Default)
	Date Format	Month/Day/Year Day/Month/Year
	Data Base Century	1900
Set Exam Number	✓ Exam Number (1-49999)	
	Diag Exam Number (50000-65535)	
Set Disk Limit	Set UDDM (60-100%)	
DASM	✓ Install DASM?	[] No [] Yes
	✓ DASM Interpolation	Linear Cubic
	✓ Laser Camera Type	3M952 3M XL or HQ 3M DryView AFGA MIN KODAK DUPONT FUJI KONICA
VCR Setup	✓ VCR Format	NTSC PAL
HIS/RIS – MR Host	Hospital Name	(Same as in "System Configure" tab)
	Suite ID	(Same as in "System Configure" tab)
	Host ID	(Same as in "System Configure" tab)
	✓ Local AE Title [Hostname] (1-16 Chars)	
	Local IP Address [Host IP]	(Same as in "Network" tab)
	✓ Local Port Number [Host Port] (0-65535)	
	Netmask	(Same as in "Network" tab)
	✓ DICOM Network Timeout	Slow (300secs) Medium (200secs) Fast (30secs)

TAB	ENTRY (✓ - Vital Info)	VALUE
	✓ Default Router IP Address	
	✓ Station Name (1-8 Chars)	
HIS/RIS – Modality Worklist	Current Patient Location	☐ No ☐ Yes
	Special Needs	☐ No ☐ Yes
	Pregnancy Status	☐ No ☐ Yes
	Referring Physician	☐ No ☐ Yes
	Request Contrast Agent	☐ No ☐ Yes
	Contrast Allergy	☐ No ☐ Yes
	Medical Alert	☐ No ☐ Yes
	Additional Patient History	☐ No ☐ Yes
	✓ MLW AE Title (1-16 Chars)	
	✓ MWL IP Address	
	✓ MWL Port Number (0-65535)	
	✓ Patient Name Delimiter	☐ No ☐ Yes
HIS/RIS – Modality PPS	✓ PPS AE Title (1-16 Chars)	
	✓ PPS IP Address	
	✓ PPS Port Number (0-65535)	
Set Time/Date	Time Zone	
--	Modem Type (Not in Install tab; look at modem)	

Note 1: In the **Hardware Configure** tab, the CERD choices are as follows:

- **CERD 1/4 Ch 62.5 62.5 62.5 62.5** – for single 62.5 KHz, or four 62.5 KHz Receivers
- **CERD 1/4 Ch 125 125 125 125** – for single 125 KHz, or four 125 KHz Receivers
- **CERD 4 Ch 125 62.5 62.5 62.5** – for 125 KHz Receiver 1, and 62.5 KHz for other Receivers
- **CERD 4 Ch 125 125 62.5 62.5** – for 125 KHz Receivers 1, 2, and 62.5 KHz for Receivers 3, 4
- **CERD 4 Ch 125 125 125 62.5** – for 125 KHz Receivers 1, 2, 3 and 62.5 KHz Receiver 4
- **UCERD 1/4 Ch 125 125 125 125** – for single UCERD, or 4 channel UCERD.

If your system configuration does not fit any of the selections above (e.g. 62.5 KHz for Receiver 1, and 125 KHz for any other Receiver), move the higher bandwidth Receiver(s) to the lower slots beginning with Receiver 1; this will put the higher BW on the most used Receivers. Then select the entry that matches your final configuration.

Check (📁) the Options installed at this Site on the System Options Table 1-2 below. After loading software and doing Restore Info, verify that all the options are still available to the customers. If any keys are missing, reactivate them using the option key floppy, or MOD.

TABLE 1-2
SYSTEM OPTIONS DATA TABLE

SITE OPTIONS	SITE OPTIONS
Echo Planar Imaging	Tagging
Fast Gradient Echo	Sgdperf
Cine	DW-EPIEZ
Fast Spin Echo & FLAIR	iDrive Pro
Time of Flight	iDrive
Phase Contrast Vascular Imaging	SmartPrep2000 Upgrade
Research	Probe2000 Upgrade
Proactive Service	Multi-Nuclear Spectroscopy
Research PSD	Functool
	Functool CSI
	Functool DTI
Spectroscopy/PROBE	Voxtool
Videoconferencing	Interavascular Imaging
SGD_EchoSpeed	Clariview
DW EPI	
FLAIR EPI	Spectro Analysis GE
SPECIAL	Performed Procedure Step
Smart Prep	High Speed
SSFSE	Below Options available on ASP2 Release or later software.
Three Plane Localizer	fgret
HIS RIS (Modality Worklist)	fgret Real Time
E3D TOF	Spiral Hi-Res
FSE_XL	Spiral Real Time
Bloodsupp	iDrive Pro Plus
Fastcine	Probe 3D Brain
	Ultrashort TR
	T2 Breathhold
	SSFSE MRCP
	T1 Breathhold
	Line Scan Diffusion Imaging
	Image Filter
	ACGD Plus
	Probe 3D Prostrate
	Fluro Triggered MRA
	MRCP3
	PSIR
	Dymanic R1
	Fiesta 2D
	Fiesta 3D

	Delayed Enhancement
	Asset
	ProbePress
	CREATIV
	Navigator
	Diffusion Tensor
	3DFRFSE
	Asset Plus
	Pomp

1-4 SaveInfo and Saving System Information Entered In The Install GUI

1-4-1 CheckINFO

1. Insert SaveINFO MOD into the MOD drive.
2. Obtain CheckINFO CD and load onto system per the following instructions.
(Part number is **5110493**.)
3. Open a C-shell.
4. Login as root by typing: **su**. The password is **operator**.
5. Insert the CD into the drive and install the SaveInfo MOD into the MOD drive.
6. Mount the CD by typing **mount /mnt <Enter>**.
7. After successfully mounting the drive type **cd /mnt <Enter>**.
8. Copy file to system by typing: **./install <Enter>**.
9. The executable should be loaded into directory **/usr/local/bin**.
10. Change directory: **cd /usr/local/bin <Enter>**.
11. Run the executable: **checkINFO <Enter>**.

The program will inform you if there are issues with the "saveInfo" data on the system. Possible problems are nonstandard characters placed into files or unusually large files.

Typical response is:

```
CalDir files are present (checking files on hard disk)
Config files are present (checking files on hard disk)
Protocol files are present (checking files on hard disk)
Option Keys are present (checking files on hard disk)
```

If issues are found, a message will display informing the operator of files that need to be moved in order for the SaveInfo to properly archive to MOD.

The list of files to be moved / corrected to make SaveInfo work properly are:

/usr/g/protocol/site/head/ noggin scan

Discuss issue with customer, and attempt fix. The corrupted file(s) can be moved to directory **/usr/g/insite/data** for temporary storage.

The executable will not check the MOD if issues are found on the system disk.

The executable will then check the SaveInfo on the MOD.

Typical response is:

```
CalDir files are present (checking files on MOD)
Config files are present (checking files on MOD)
Protocol files are present (checking files on MOD)
Option Keys are present (checking files on MOD)
```

Unmount the CD by typing **umount /mnt**

1-4-2 SaveInfo

A SaveInfo must be done.

1. Insert the SaveInfo MOD or a new MOD into the drive.
2. Click on **[Save Info]** from the Install menu. The system opens the window called Xexec. (If the MOD has not been used, you will see a message regarding the use of the MOD for "Raw Data Storage"; choose **[Overwrite]**).
3. When asked: Do you really want to save configuration files to the Info MOD? [y|n], type: **y <Enter>**.
4. When SaveInfo is complete, you can try the following command to give you an indication that the SaveInfo was successful. Open a C Shell on the Service Desktop. Login as **root** by typing: **su root**, then enter password **operator**.

Type the following:

tar vtf /MOD_DRIVE <Enter> (there is a space between vtf and / character)

*Hundreds of files that were backed up on the MOD should scroll by at this point. This will be a good indicator that the SaveINFO was successful. Note: to view the files a page at a time you can type: **tar vtf /MOD_DRIVE|more <Enter>** (the vertical character "pipe" followed by the word "more" will display a page of output at a time; use the space bar to continue paging).*

Also as an indication that site protocols were saved successfully, you should see the something similar to the following (protocol names will not match, this just an example)

```
rw-rw-rw-  0/0              0 Jul 28 14:26 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/chest/.MrProtocol.chest.cfg
rwxrwxr-x  0/0  dir              Aug  7 10:17 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/
rw-rw-rw-  0/0              9472 Jun 28 13:20 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/.MrProtocol.head.cfg
rwxr-xr-x  64/32116         298496 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/1.0T_destroyer_head_A
rwxr-xr-x  64/32116         587898 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/Sr20PerformSpecs
rwxr-xr-x  64/32116         298602 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/destroyer_head_A
rwxr-xr-x  64/32116         132142 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2d_thick
rwxr-xr-x  64/32116         22059 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2dfast_cmon
rwxr-xr-x  64/32116         16571 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2dfast_fcOFF
rwxr-xr-x  64/32116         16572 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2dpc_fcOFF
rwxr-xr-x  64/32116         16592 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2dtof_fcOFF
rwxr-xr-x  64/32116         27806 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_3d_thick
.....
```

If site protocols are missing, please call OLC for support and do not proceed.

- At top left corner of the Install GUI, select **<File>** then **<Save GI Configuration to MOD>**.

Note

The "Save GI Configuration to MOD" creates a copy of the information already entered in the GUI tabs for later use to update the tabs during a software reload. The **<Save GI Configuration to MOD>** and SaveInfo can be saved on the same side of the Info MOD. A separate partition is built for each. It is important to note that the "Save GI" configuration does not create a complete SaveInfo disk.

- Remove the MOD and write release version, the system name, and the date on it.

1-4-3 Record Site Coils

As part of the software load, all GE coils will be included in the Site Coils List. You will want to remove the GE coils that your customer does not have after the load, so record the site coils currently listed in the Coil Config File as follows.

- From the Service Desktop, select **[Utilities]**, then **Config File Manager**. Select **Coil Config File**.
- Record all coils below that are listed under **Site Coils**.

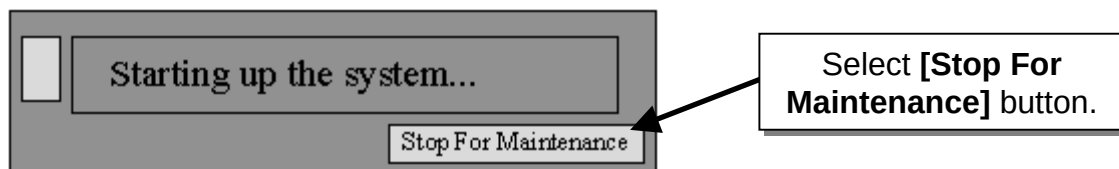
2- HOST OPERATING SYSTEM SOFTWARE LOAD

Initial Conditions:

- All images archived (Section 1-2).
- System information recorded (Section 1-3); SaveInfo performed (Section 1-4).
- System disconnected from Hospital Network.
- DASM disconnected from Host.

Procedure:

- If the system is using two **9** Gigabyte disk drives, power down the computer and remove the middle disk drive. Failure to remove the middle drive at this time will result in a failed install. This drive will be reinstalled and configured at the end of the software install.
- If the system is not up and running, switch on power at this time and continue with step 2. If the system is already running, go to the Service Desktop and select **[Shutdown]**. When prompted, click on the **[Restart Button]**.
- Stop the reboot process; click on the **[Stop For Maintenance]** button when it displays.



4. Click on the **[Enter Command Monitor]** button when it becomes available.
5. Insert the **IRIX 6.5** CD labeled "**Boot/Update**" into the CDROM Drive on the workstation (right side). On the command line type the following (*user responses are bolded text*):

WARNING Revision 9.1 software can only be loaded onto Octane Computers

For SGI Octane computer:

>> **boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 --x** <Enter> (Note two minus signs)

Note

The rest of the install is for all SGI computer types.

```
fx: "device-name" = (dksc) <Enter>
.
fx: ctrlr# = (0) <Enter>
.
fx: drive# = (1) <Enter>
.
fx> r/ro <Enter>
.
fx/Repartition/RootDrive: Type of Data Partition = (xfs) <Enter>
.
fx> continue? yes <Enter>
.
fx> l/sy <Enter> (This is a lower case l , /, lower case s, and lower case y.)
.
fx> exit <Enter>
```

6. The system will reboot. Select **[Install System Software]** button. Choose, **[Local CD-ROM]** (the default), select **[Install]**. Then select the **[Continue]** button.

Note

If you are working with new or never used (Unformatted) disk drives, you will be told by the system during this reboot that the system was unable to mount the "/ root" file system. Without a format, the file system (Directory "/ root" being one of file structures) cannot be built. The system will stop with the statement: "press Enter to invoke C shell csh:". Type the following:

1. ...press Enter to invoke C shell csh: <Enter>
2. # **mkfs /dev/dsk/dks0d1s0** (This is /dev/dsk/dks zero d one s zero.)
(The file system will be built at this time.)
3. # **exit** <Enter>
4. **Reset the computer.** The computer will begin to reboot.
5. **Restart this procedure at Step 2, Section 2.**

Note

Instead of seeing **dks0d1s0** below, you might see **realroot**. Select **yes <enter>** to continue, this is not a problem.

7. Make new file system on /dev/dsk/dks0d1s0 [yes/no/sh/help]: **yes<Enter>**
About to remake (mkfs) file system on: /dev/dsk/dks0d1s0.
This will destroy all data on disk partition: /dev/dsk/dks0d1s0.
8. Are you sure? [y/n] (n): **y<Enter>**
9. Block size of filesystem 512 or 4096 bytes? **4096<Enter>**
Doing: mkfs -b size=4096 /dev/dsk/dks0d1s0
meta-data=/dev/rdisk/dks0d1s0 isize=256 agcount=8, agsize=128612 blks
data = bsize=4096 blocks=1028895, imaxpct=25
log =internal log bsize=4096 blocks=1000
realtime =none bsize=65536 blocks=0, rtextents=0
Trying again to mount /dev/dsk/dks0d1s0 on /root.
10. Continue as follows:
inst> **sh <Enter>**
.
mount /CDROM <Enter>
.
/CDROM/mr_prep <Enter>
.

Note

During the software load, there will be much screen output and many pauses of 10 to 30 seconds, no entry is required at these pauses. Approximate wait times are listed for any lengthy pauses, wait for the prompts as detailed in the following steps.

Note

If you receive the following error at this time, verify DASM is disconnected from the host. **DKSC..... Ran out of disk space**

11. The system will run for about 4 minutes and then will prompt with:

"Please insert the First OS Disk (Pa) and then press <enter>".

Put the OS CDROM Disk 1 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.
12. After an additional 3-6 minutes, the system will again prompt with:

"Please Insert the Second OS Disk (Pb), then press <enter>"

Put the OS CDROM Disk 2 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

Loading the rest of the Operating System off the second disk will take about 30 minutes and will not require any user intervention.

13. When finished, the system will stop at the “#” prompt. Type the following to continue.

```
.  
# exit <Enter> (to exit back to the original inst> prompt.)  
.  
.  
Inst> exit <Enter>      there is a delay of about 30 seconds, then:  
.  
Automatically reconfiguring the operating system..  
.  
.  
.....Ready to restart... y <Enter> (Restarts the system)  
.
```

The system will finish some processing and reboot. The OS load portion of the Software Install is complete.

Important: If at any time during the OS load a problem occurs that causes you to stop the install, correct the problem then reset or power cycle the computer and begin the OS install again from the start. This is necessary to insure a complete load. If this does not fix your problem it may be necessary to do a low level format on the system drives. Refer to Appendix C for this process.

14. Continue to Section 3 of this document for the MR_Apps software load.

3- MR_APPS SOFTWARE LOAD

1. When the system is finished rebooting, log in as **root**. There is no password at this time. You will first see a startup box that will asks for the MrpApps CDROM. See Illustration 3-1.



GUIDED INSTALL LOAD SCREEN

ILLUSTRATION 3-1

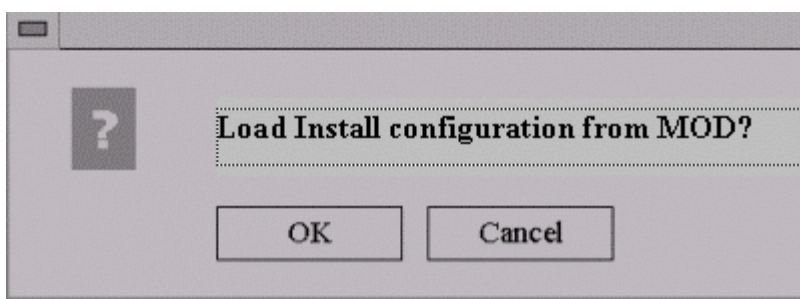
2. Remove any CD from the CDROM drive and insert the MrpApps CDROM into the CDROM drive. Wait 20 seconds or until the amber light on the CDROM drive stops blinking and click on the **<Load>** button. The Guided Install will load and start.

Note

Help documentation is built into the Guided Install GUI. Push the **<Help>** button on the tab you need information for in the GUI. When the GUI first starts, press the **<What's New>** button for information on using the GUI.

3. On initial startup, a popup box will display over the Guided Install interface describing the tool and its use. After reading the information push the **<OK>** button to continue.
4. Next, a popup box (see Illustration 3-2) will display over the Guided Install interface.

Important: The last Install GUI configuration can only be installed if it was previously saved using the **<Save GI Configuration to MOD>** option per Section 1-4-2, Save GI configurations.



INSTALL CONFIGURATION POPUP SCREEN

ILLUSTRATION 3-2

5. Insert the SaveInfo MOD (created in Section 1-4-3, Save GI Configurations) then push the **<OK>** button to reinstall your last GUI install configuration values. *(The normal SaveInfo files are not restored at this point in the software load. They will be restored in step 9.)* You will verify the correct values were restored by checking each tab in the next step.

Note

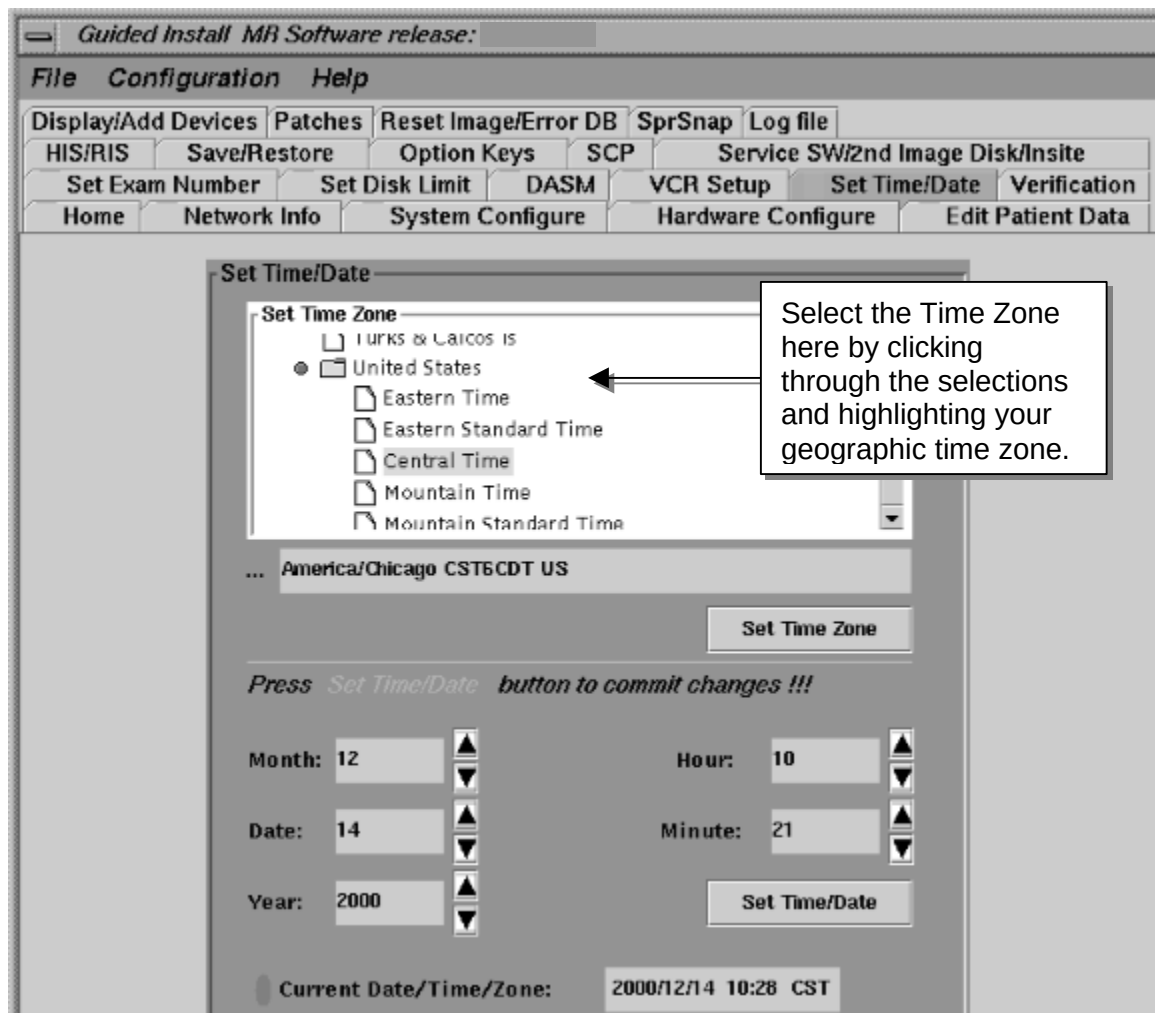
If you did not perform Section 1-4-2, Save GI Configurations (for example, sites upgrading from 8.2.5 software), push the **<Cancel>** button to continue manually with the install.

6. The Guided Install home screen is displayed. See Illustration 3-3. It is now OK to go through the tabs using the mouse.



GUIDED INSTALL - HOME SCREEN
ILLUSTRATION 3-3

7. Press the **<Next Tab>** button to move through active tabs from “**Network Info**” to “**Set Time/Date**”. Fill out all empty entries or make any updates as appropriate. This information should have been saved in Table 1-1 of this document. When you reach the “Set Time/Date” tab, go to the “Set Time Zone” box and use the selections to set the time zone your system is in. See Illustration 3-4.

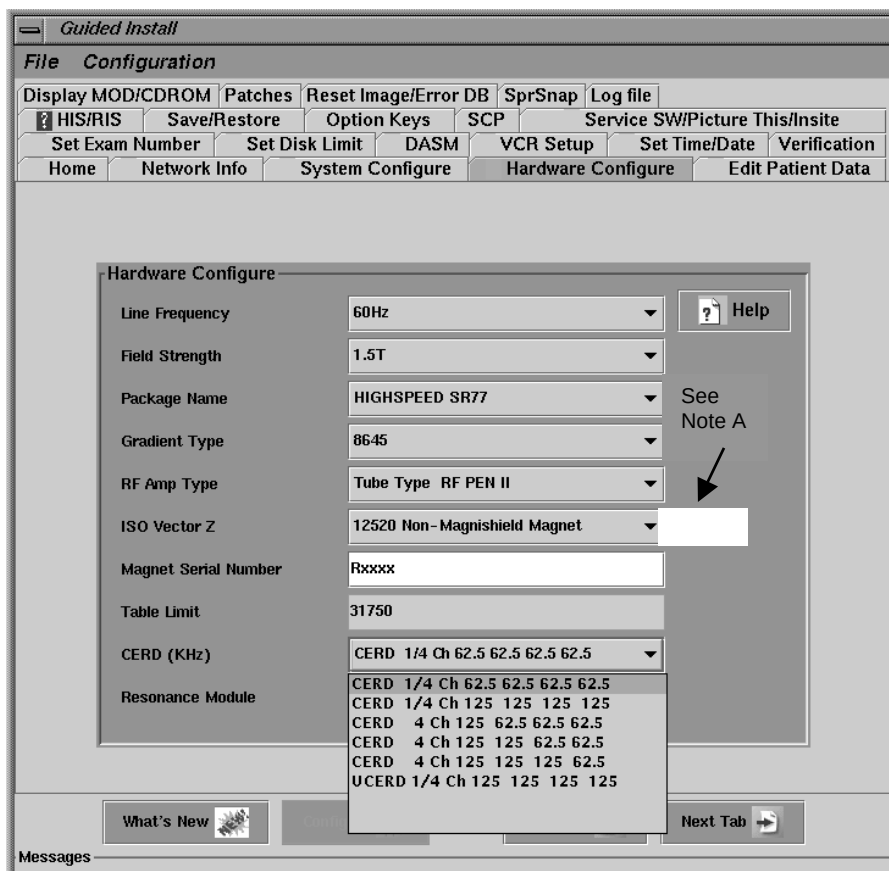


GUIDED INSTALL - SET TIME/DATE SCREEN
ILLUSTRATION 3-4

8. In the **Hardware Configure** tab, there is an entry for CERD that is difficult to understand. See Illustration 3-5. Refer to the Receiver Bandwidth entries you entered in Table 2-1 and choose the appropriate selection as follows for CERD (KHz):
- **CERD 1/4 Ch 62.5 62.5 62.5 62.5** – for single 62.5 KHz, or four 62.5 KHz Receivers
 - **CERD 1/4 Ch 125 125 125 125** – for single 125 KHz, or four 125 KHz Receivers
 - **CERD 4 Ch 125 62.5 62.5 62.5** – for 125 KHz in slot 1, and 62.5 KHz for other Receivers
 - **CERD 4 Ch 125 125 62.5 62.5** – for 125 KHz in slots 1, 2, and 62.5 KHz in slots 3, 4
 - **CERD 4 Ch 125 125 125 62.5** – for 125 KHz in slots 1, 2, 3 and 62.5 KHz in slot 4
 - **UCERD 1/4 Ch 125 125 125 125** – for single UCERD, or 4 channel UCERD. This is to support the new 8.3 configuration. There are no UCERD's on Release 8.2.5 systems.

Note

If your system configuration does not fit any of the selections above (e.g. 62.5 KHz for Receiver 1, and 125 KHz for any other Receiver), move the higher bandwidth Receiver(s) to the lower slots beginning with Receiver 1; this will put the higher BW on the most used Receivers. Then select the entry that matches your final configuration.



GUIDED INSTALL – CERD CONFIGURATION
ILLUSTRATION 3-5

Note A

This field is the for the calibrated ISO Vector Z. It will be blank on a normal software Load and will be filled in during the RestoreInfo process.

9. Proceed to the **Verification** tab. Once in the Verification tab, look at the pass or fail status of the entries. If any entry has a failed status, the **<Install Software>** button will not be active. Failed entries will be due to missing entries, or entries not in the correct format. Go to the tab with the error and correct it, then push the **<Install>** button to open the Verification tab again. If all entries pass verification at this point, push the **<Install Software>** button again to begin the MR Apps load. The MR Apps load will take 20-30 minutes, depending on the machine being loaded.

Note

It is not necessary that any one tab be filled out in any particular order. Just all required configuration tabs must be filled in or the install will not continue. The Verification Tab will list any formatting errors as well as current settings. See Illustration 3-6 for an example of a failure in the Verification Tab in networking (in the example, the Host IP Address is an incorrect format).

Verification Failed!

Status	Parameters	Value
	Time Zone	America/ChicagoCST6CDTUS
	Field Strength	1.5T
	Package Name	ECHOSPEED
	Gradient Type	8651
	Resonance Module	CRM
	Magnet Serial Number	DXXXX
	Table Limit	31750
	Line Frequency	60Hz
	RF Amp Type	Tube Type
	Governing Body	fda2
	CERD (KHz)	CERD 7/4
	ISO Vector Z	8981 CX /
	Isocenter Vector along Z-axis	8981
Fail	Host IP Address	3.7.25.
	Netmask	0xffffc00
	Hostname	t9
	Domain Name	mrceg
	YP(NIS) Service	On
	IWS PC IP Address	3.7.27.9
	Magnet Monitor IP Address	1.1.1.0
	FE Laptop IP Address	192.168.4.2
	TPS Subnet	216.33.27
	Default Router IP Address	
	Power Monitor PC IP Address	
	Hospital Name	t9

Errors are shown in red on color monitors, or light gray in Monochrome monitors.

Error details are shown in the message area at the bottom of the Install GUI. This failure is due to an incorrect IP address format. The correct format is:

xxx.xxx.xxx.xxx

The **<Help>** button on the Network Tab gives the correct format.

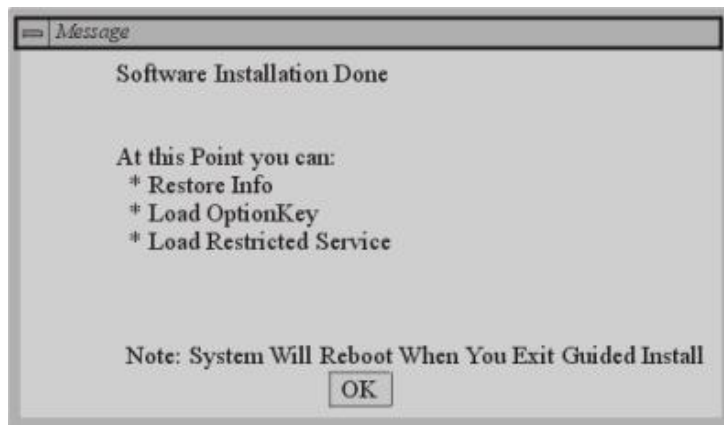
7:37:28; GUI Warning ! Invalid entry. This field cannot begin or end with a dot----- Host IP Address : 3.7.25.

Status

Enter IP Address (eg. 2.33.44.55)

GUIDED INSTALL - VERIFICATION SCREEN
ILLUSTRATION 3-6

10. Once the MR Apps load is done the popup in Illustration 3-7 is displayed. Click on the **<OK>** button to close the popup.



GUIDED INSTALL – SW DONE SCREEN
ILLUSTRATION 3-7

Note

Do not load restricted software at this time.

11. At this point the GUI will change from the Install phase to the Configure phase. If a SaveInfo info MOD is available for this site, proceed to the **Save/Restore** tab and select **[Restore Information]**. If no SaveInfo MOD is available, go to each tab in the Install GUI and fill in or update all system information manually. A help button is available on each tab, which will open a popup box with information on the required data and its format for that page.
12. A popup will ask if you want to restore all files or selectively restore, see Illustration 3-8. Select **Restore all files automatically**, then **[Continue]**.



RESTORE OPTION POPUP
ILLUSTRATION 3-8

Note

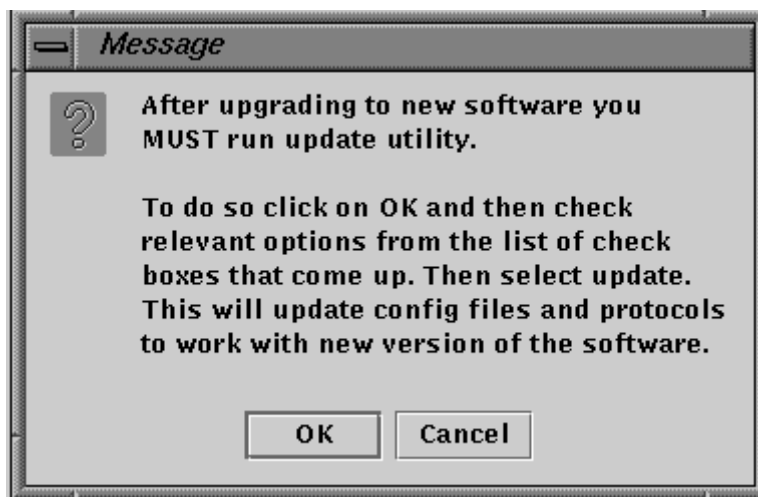
If RestoreInfo fails, the CDROM reader will automatically eject the media. Exit install and allow system to reboot. When the system is back up, re-enter *Guided Install* and attempt to RestoreInfo again.

13. A “RestoreINFO in Progress” window will scroll filenames as they are being restored and then a popup will ask if you want to specifically restore the MRconfig file. See Illustration 3-9. Select **[Yes]**.



MRCONFIG FILE RESTORE POPUP
ILLUSTRATION 3-9

14. When the RestoreInfo is finished a popup will open telling the installer that to run the update utility. See Illustration 3-10.



GUIDED INSTALL - UPDATE UTILITY MESSAGE
ILLUSTRATION 3-10

15. Push the **<OK>** to continue. The Restore Manager menu opens (see Illustration 3-12).



GUIDED INSTALL – RESTORE UTILITY
ILLUSTRATION 3-11

Make sure to check the following:

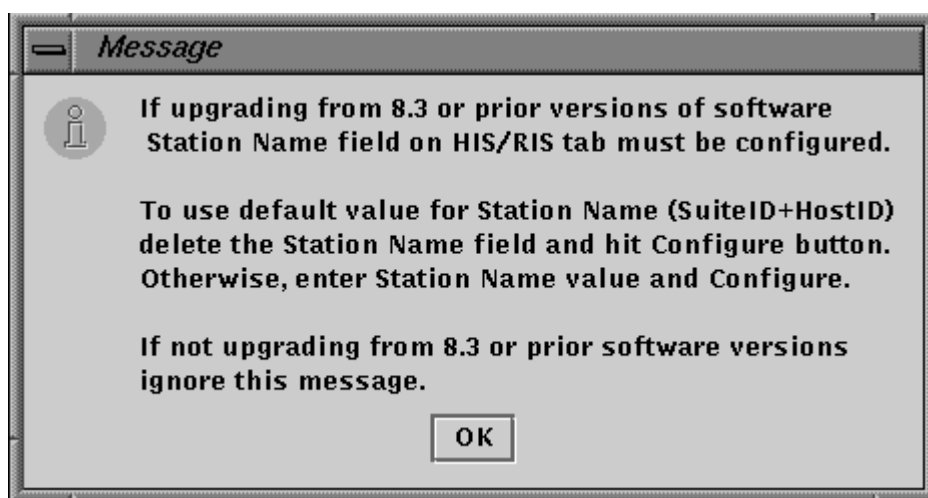
- a. CoilConfig.cfg
- b. MRconfig.cfg
- c. GradientConfig.cfg
- d. Protocols

16. Click on **<Update>** button when update is finished Click **<FILE>** then **<Quit>**.

A logfile of changes are made by the Restore Manager and is stored at:
/w/config/restore.log

17. Select **[OK]**.

18. When the message shown in Illustration 3-12 displays, select [OK].



8.3 UPGRADE MESSAGE
ILLUSTRATION 3-12

IMPORTANT

The following Prose Option Key should be removed from the system and reloaded from MOD to ensure proper operation. Failure to do this will remove the PACC button from the Browser.

IMPORTANT

The following Cardiac Option Keys should be removed from the system and reloaded from MOD when upgrading to revision 9.1 software. Failure to do this will prevent the options from working properly.

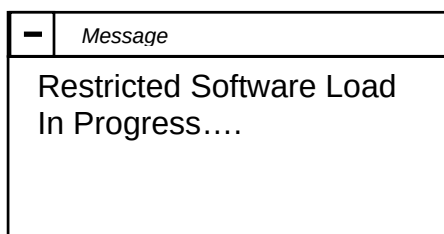
- Any Fiesta Key
- Delayed Enhancement
- Diffusion Tensor
- Navigator
- IVI

19. Use the cursor to select <File> and then <Quit> at the top left of the Install GUI. When the GUI shuts down, the system will reboot and all the changes made during the restore process will be set.

20. After the system has rebooted, log in as **root**. Password is **operator**

21. Select [Yes] to start the Install GUI.

22. Go to the **ServiceSW/2nd Image Disk/InSite** tab and install service software. (*Restricted Service Software is only available for GE use; Advanced Service Software is only available to sites with a valid Advanced Service Package Limited License.*) GEMS Field Engineers loading GEMS Restricted Service Software should also configure InSite Interactive at this time (see Illustration 3-13). Follow the instructions given in popup windows as presented. For a more complete discussion on configuring the InSite package, see Appendix E of this document.



The screenshot shows the 'IIPConfig' window titled 'InSite Interactive Platform Configuration'. It has several tabs: 'Information', 'ProDiags', 'Health Page', 'Device Connection' (which is selected), and 'InSite Checkout'. Below the tabs, the title 'Connection Device Configuration' is displayed, followed by 'Last Update: Fri Aug 13 15:50:42 CDT 1999'. A dropdown menu for 'Device Connection Type' is set to 'Modem'. A text box contains instructions: 'Select the InSite Connection Type above, or select the Modem settings given below and Press APPLY - To update the modem and modem configuration files with model and country defaults. CUSTOM - For other modem settings.' Below this, there are several configuration fields: 'Dial-out Prefix' (set to '8'), 'Dialing mode' (set to 'Tone'), 'Modem Type' (set to 'USRobotics Courier V.34 or V.Everything'), 'Country' (set to 'Default - All Others'), 'CPU Serial Port Name' (set to '/dev/ttyf1'), and 'CPU Serial Port Speed' (set to '115200'). At the bottom right are 'APPLY' and 'CUSTOM >>' buttons. A status bar at the bottom left says 'Status: IIP Configuration Tool Ready for Input...' and a button labeled 'EXIT' is on the bottom right.

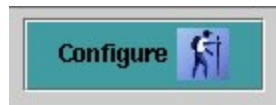
Note

When configuring the System Health page of **Configure InSite Menu**, you are able to enter multiple email address where reports are sent. The instructions on that tab are incomplete. You must push the **<Enter>** key after every email address entered.

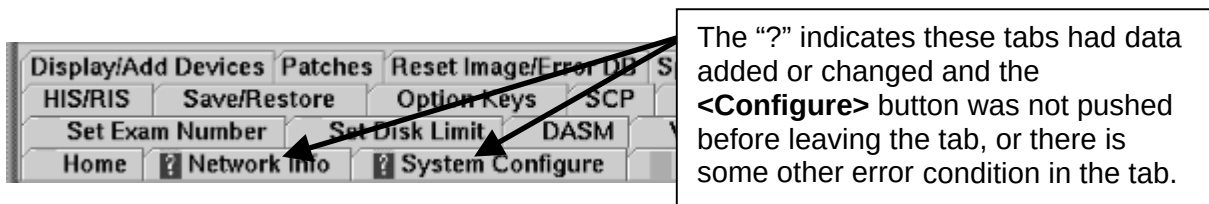
Note

Release 8.3 version M4 and later software comes with two software MOD keys that must always be reloaded. These two MOD's (two sides on one MOD and one side on the other MOD to load) contain the LX Tools software that is shipped with every system.

23. Turn YP on only if the site's external network requires it. If YP is turned on, the system will take longer to boot (2 to 4 minutes). Be aware there is a small delay that may not be seen on the sites.
24. After information as been restored from MOD or changed in a tab, and configuration updates have been run, push the **<Configure>** button on the bottom of the tab to apply the configuration.



If the **<Configure>** button is not pushed to commit changes to a tab, or an error is present in a tab, a question mark will be seen on the tab, as shown in Illustration 3-13.



CONFIGURATION PROBLEM EXAMPLE
ILLUSTRATION 3-13

25. Perform a SaveInfo to save the latest changes by inserting the site SaveInfo or blank MOD into the drive then going to the Save/Restore Tab and pressing the [Save Information] button.
26. Insert the your normal SaveInfo MOD or a new MOD into the drive if not already in the drive.
27. At top left corner of the Install GUI, select **<File>** then **<Save GI Configuration to MOD>**. This will save information that has been entered in all the edit fields in the GUI.

Note

The “save to MOD” process does not create a complete SaveInfo disk. It only creates a copy of the information already entered in the GUI tabs for later use to update the GUI tabs during a new install. (*SaveInfo uses the Save/Restore tab.*) The <Save GI Configuration to MOD> and SaveInfo can be saved on the same MOD. A separate partition is built for each.

28. Exit the install GUI by selecting **<File>** then **<Quit>** from the top of the interface. If any error conditions still exist, you will be warned that no changes will be made. If no error conditions are found, the GUI will save user parameters.
29. Reconnect the following hardware at this time if it was disconnected at the beginning of the install:
 - Network Interface
 - DASM
30. If applicable run the **Install DASM** command by selecting the DASM option from Guided Install then select **Install DASM**.
31. If the system was using two 9 Gigabyte disk drives the second drive should be reinstalled at this time. If not, go to step 30.
 - A. Log out of root and shutdown the computer and remove power.
 - B. Reinstall the second disk drive into the middle drive slot.
 - C. Power up the system and log into root and start the configuration GUI. See Illustration 3-14.
 - D. Go to the **Service SW/2nd Image Disk/Insite** tab and push the **[Install 2nd Image Disk]** button. The disk will be configured in within seconds. See Illustration 3-14

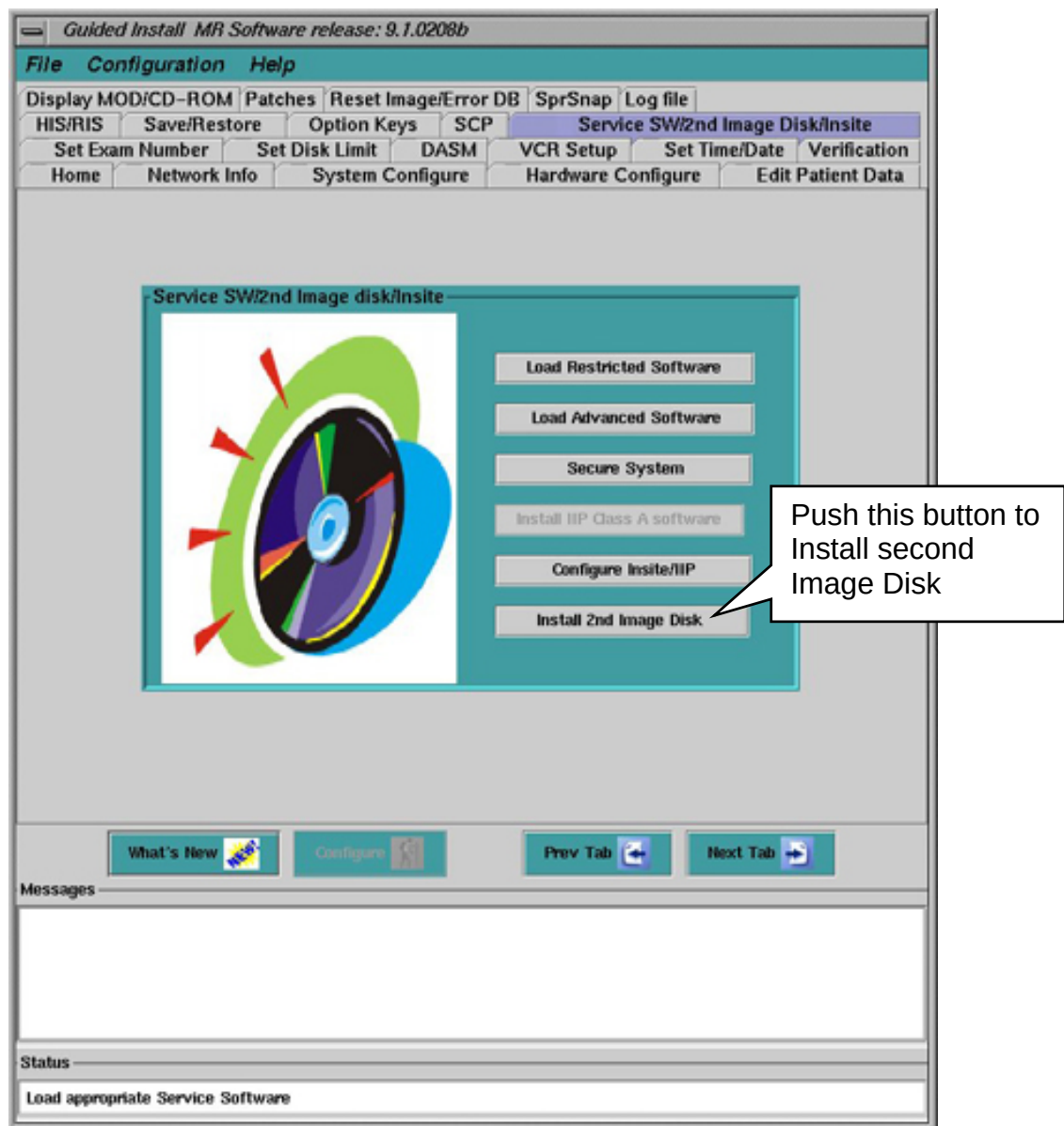


ILLUSTRATION 3-14
INSTALL 2ND IMAGE DISK BUTTON

32. Log out of root and reboot the system.

4- VCR INTERFACE OPTION

If your system does **not** have the VCR I/F Option proceed to Section 5. Systems that have the VCR Interface Option will have to follow the steps in this section to load the video driver because it is not loaded with the operating software load. Follow these steps to install the video driver software:

1. From the log in screen, log in as **root**. Password is **operator**. You will be asked once entering the root desktop if you want to start the Install GUI. Select **[No]**.
1. Insert the IRIX 6.5 Boot/Update CDROM into the CDROM drive. For 9.1 software insert the Printer/VCR Options CDROM into the CDROM drive.
2. In the winterm window type: **mediad <Enter>** (to mount the CDROM drive).
3. Type: **cd /CDROM/opt <Enter>**
4. Type: **./vidsetup <Enter>**

This starts the install script and a lot of screen output will scroll by in the window, and it will be apparent that file installs/removals are taking place. Then the last output to be displayed is below:

```
Reboot to start using the reconfigured kernel.
*****
**                                                     **
**                                                     **
**      Remember to reboot the system before          **
**      attempting to use the video interface          **
**                                                     **
*****
{sdcc@name}
```

5. Type the following to verify the video driver was loaded:
showprods -D1 <Enter>

examine the products that are loaded and ensure the following is listed:

evo <date installed> OCTANE Personal Video Execution Environ, 6.5.4m

6. Press the spacebar until you get a prompt back.
7. Type: **mediad -k <Enter>** (to release the CDROM device).
8. Remove the Boot/Update CDROM.

Note

Do not reboot the system at this point to activate the video driver. The system will be rebooted after loading the patch software in the next section.

5- MONITOR GAMMA VALUE RESET

After a complete software load has finished, you will need to set gamma values for the system monitor to insure image quality and camera to monitor compatibility. The complexity of the gamma setup has required the need to consolidate adjustment document. To complete this step, chose the appropriate document from the Service Methods CD ROM for your system.

- For color and black and white use the procedure [Models MDW321 and MDW321+ Monitor Adjustments].
- For NEC LCD monitors use the procedure [NEC 2000, 2010, 2010X Host LCD Color Monitor Calibration].
- For Eizo L660 LCD monitors use the procedure [EIZO L660-Host LCD Color Monitor Adjustment].
- For NEC 1850X LCD monitors use the procedure [NEC 1850X – Host LCD Color Monitor Calibration].

If you do not see your monitor, view the list of document on the service CD ROM under Operator WorkSpace for updates. If no procedure is available for your monitor and it is a GEMS supplied monitor, contact support.

6- RECONFIGURING THE SYSTEM USING THE INSTALL GUI

If this is a new/upgraded Signa installation or if the system just needs to be re-configured between software loads, do the following to start the Configuration GUI:

1. With Signa up and running, go to the service desktop and push the **<Install>** button on the toolbar. A C-Shell will open and ask for root password ("**operator**" **<Enter>** is the default). The Install GUI will load and start.
2. Help documentation is built into the GUI. Push the **<Help>** button on the tab you need information for in the GUI. Push the **<Next>** button to move through tabs in order. Modify entries to match your system as required. When satisfied, push the **<Configure>** button at the bottom of the tab page or wait until all tabs have been modified and go to the "**Verification**" tab. Review changes made are correct, then push the **<Configure Tab(s)>** button to update the system (see Illustration 4-1).



VERIFICATION MENU – CONFIGURE TABS
ILLUSTRATION 4-1

3. Before exiting the GUI, insert the SaveInfo MOD or new MOD into the drive. Go to the top left corner of the Install GUI and select **<File>** and then **<Save GI Configuration to MOD>**. This process does not create a SaveInfo disk. It just creates a copy of the information already entered in the GUI tabs for use in the next software install.
4. Exit the install GUI by selecting **<File>** then **<Quit>**. If any error conditions still exist, you will be warned that no changes will be made before exiting.

Note

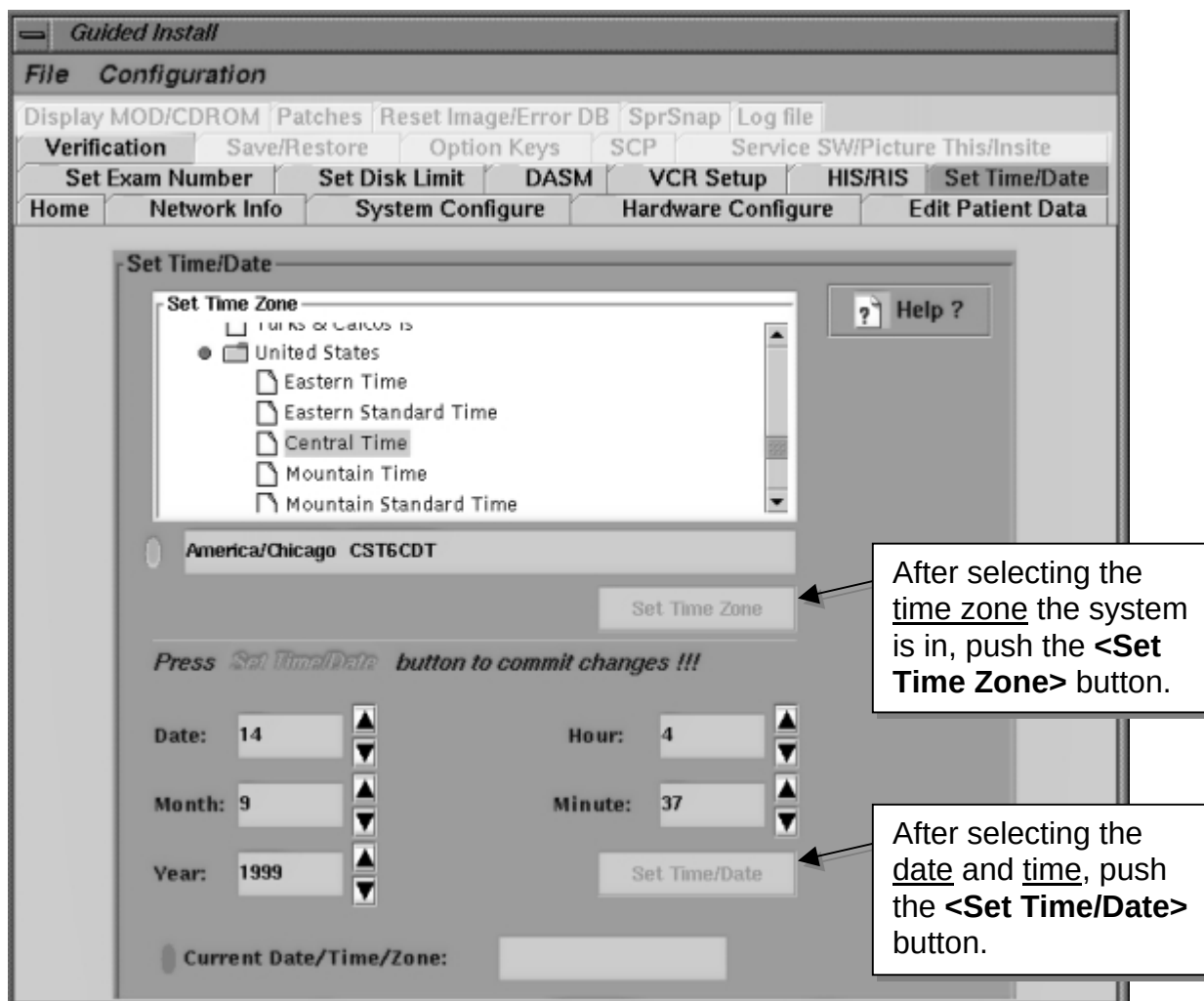
The GUI will not automatically reboot the system in configure mode. It will be up to the user to reboot after making configurations.

5. If changes were made to the system configuration, select **[System Shutdown]** and reboot the systems for changes to take effect.

7- SET TIME/DATE TAB

The **Time/Date** tab is the only tab that the Configure button will never be highlighted to set options. There are separate buttons for configuring this tab (see Illustration 7-1). If you need to re-configure the Host time and/or date at any time, do the following:

1. Select the Time Zone your system is located in, then push the **<Set Time Zone>** button.
2. Select the date and local time, then push the **<Set Time/Date>** button.



SET TIME/DATE MENU
ILLUSTRATION 7-1

APPENDIX A - HOST SOFTWARE INSTALL WORKSHEETS

Host IP Address: The Host IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

IWS PC IP Address: The IWS PC IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

FE Laptop IP Address: The Laptop IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

TPS Subnet Address: The TPS is on a subnet of the Signa LX scanner. A default address is loaded during the load from cold process and it should not be changed.

Hostname: This is the network name given to the scanner host computer.

Hospital Name: The Name of the hospital or other facility where the Horizon LX is installed. Name can include characters and/or integers up to 32 characters.

Doctors Title: The title to be used at the scanning site by the reading physician (Radiologist or Diagnostician).

Suite ID: Name of the Suite (if any) the scanner will be connected to. Leave blank otherwise.

Unique ID: The Unique System ID is a 16 alphanumeric character string assigned at the time the system is first installed. The Unique system ID is assigned by GE CARES when the first dispatch is opened to document the start of installation.

Service ID: The Service ID is a 16 alphanumeric character string assigned at the time the system is first installed. This is the same number that GE CARES uses to track financial and administrative data for the system.

Weight Unit: The weight system to be used by the scanner. Kilo Grams or US pounds.

Monitor type: The type of monitor the scanner will be using.

Windows 95 OEM Number: The Microsoft OEM Number found on the "Proof of Authenticity" for Windows 95. _____

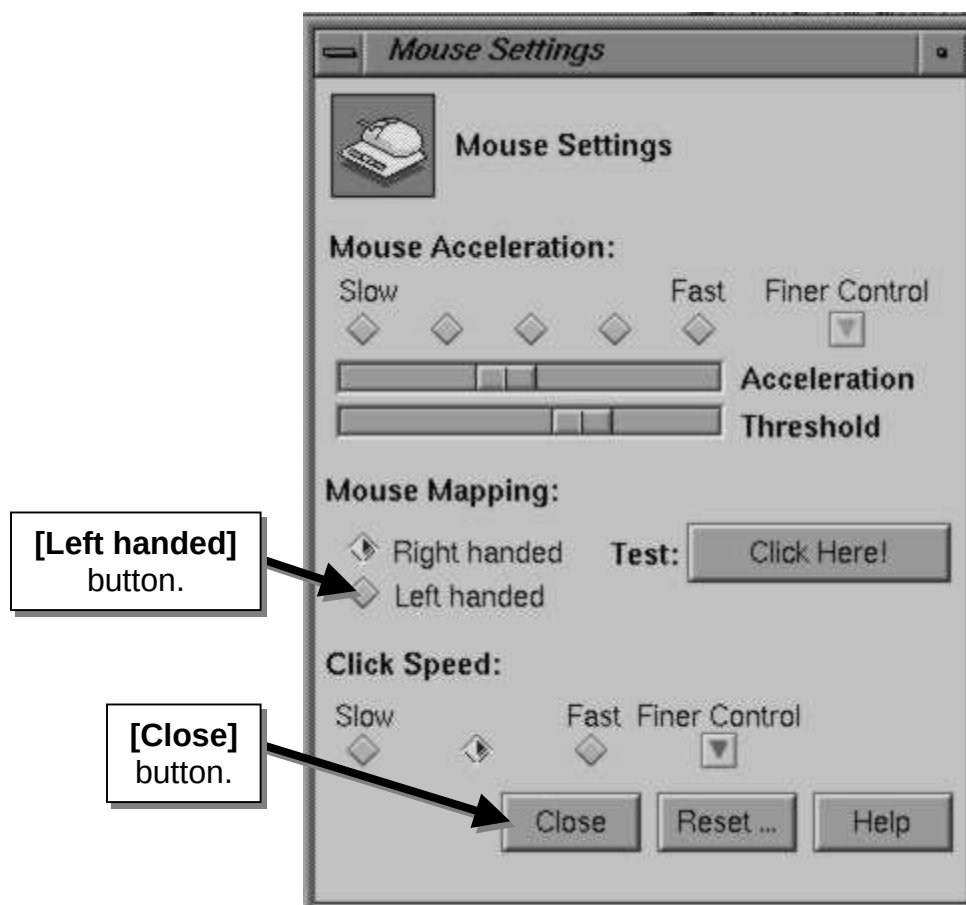
NetMask: For new installs, the Netmask should be obtained from the site network administrator if the scanner will be connected to the site's internal network. If this is a stand alone system, use the default netmask. For Release 8.3 and later, the netmask value is available from the Install GUI. For Release 8.2.5 systems, do the following to obtain the netmask value: **[System Shutdown]**, Log in as Root, type **operator <Enter>**, select **[Reconfig]** then **[Basic Network Setup]** to view the netmask.

APPENDIX B - CONFIGURING MOUSE FOR LEFT HANDED USE

The system Host mouse is default configured for right handed use. It can be configured for permanent or temporary left handed use.

B-1 Configuring Mouse For Permanent Left Handed Use

1. If Signa is up and running, you must first log out and bring the system up to the Login screen. Otherwise, turn the system on and proceed to the Login screen.
2. Log in as **root**. The system default password is "**operator**". On the root desktop, use the mouse cursor on the ToolChest and click and slide through **[Desk Top]** then **[Customize]** then **[Mouse]**.
3. Push the **[Left Handed]** button under Mouse Mapping, then push the **[Close]** Button to accept the changes. See Illustration B-1.



SET TIME/DATE MENU
ILLUSTRATION B-1

4. Log out of the root desktop and reboot the system. To change the mouse mapping back to Right handed. Repeat this section but push the **[Right Handed]** button, then reboot the system.

B-2 Configuring Mouse For Temporary Left Handed Use

Using this method will change the mouse mapping from right handed to left, but only until Signa is rebooted, when it will revert back to right handed mapping.

1. If not already at the Signa scanning level, go there. Open a command window and on the command line type: **mouse <Enter>**.
2. Push the **[Left Handed]** button under Mouse Mapping, then push the **[Close]** Button to accept the changes. (See Illustration B-1.)
3. To reconfigure the mouse back to right handed operation, repeat this section but push the **[Right Handed]** button, or reboot Signa.

APPENDIX C - DISK DRIVE LOW LEVEL FORMAT

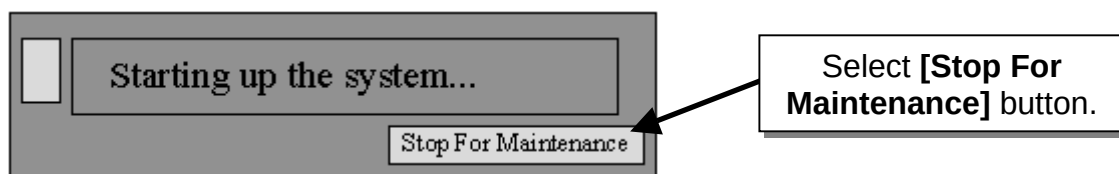
This section describes how to do a low level format of the disk drives. A low level format should only be done if loading a different Operating System (OS) or if suspected bad blocks on a disk drive are causing system problems. Depending on the system and disk drives, this procedure can last up to 1 hour per drive or as little as 20 minutes per drive.

1. Insert the **IRIX 6.5** CD labeled "**Boot/Update**" into the CDROM Drive on the workstation (right side).

Note

User responses are bolded text

2. With the system up and running, go to the service desktop and select **[Shutdown]**. If the system is not up and running, switch on power at this time.
3. When prompted, click on the **[Restart Button]**. Stop the reboot process by clicking on the **[Stop For Maintenance]** button when it displays.



4. Next, click on the **[Enter Command Monitor]** button when it becomes available. Type one of the following depending on the system your working on:

For SGI Octane computer type:

```
>> boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 --x <Enter>
```

For SGI Indigo computer, type:

```
>> boot -f dksc(0,3,8)sashARCS dksc(0,3,7)stand/fx.ARCS --x <Enter>
```

5. The rest of the install does not depend on the type of SGI computer Octane. Continue typing as follows:

fx: "device-name" = (dksc) <Enter>

fx: ctrlr# = (0) <Enter>

fx: drive# = (1) <Enter> or 2 <Enter> if doing the second drive.

fx: d/fo <Enter>

Press <Enter> to accept current parameters.

Type: **Yes** <Enter> to start the format.

6. The format can take up to 1 hour, depending on the system and disk drives that are being formatted. There will be no prompts to the user while the format is running. Be patient and do not interfere with the system. When the format is finished the system will prompt with:

Format completed successfully

.

.

fx/debug>

7. Reset or power cycle the computer to exit debug and proceed to Section 2 of this document to proceed with the OS (Operating System) load for the system.

APPENDIX D - GENERAL INFORMATION AND NICE TO KNOWS

D-1 Bore Temperature Sensor - Must Be Working!

If the bore temperature sensor is disabled, sites with *iDrive* software will have problems. Sites running the *iDrive* software must have a functional bore temperature sense circuit. The symptom for a non-functional Bore sensor will be evident when an *iDrive* scan is attempted; the prescan will complete OK, but the scan will stop immediately after the first pulse is heard in the body coil. No images will be produced and no errors will appear in the log.

If the system has *iDrive* software (as listed in *Option Key* tab of Guided Install), to verify functional bore temperature circuit, set up the scan in Table D-1 (or any other Real Time scan). Download the scan and select **[Scan]**. If prescan completes and the scan continues to run, you have a functional bore temperature circuit. If prescan completes OK, but the scan stops immediately after the first pulse, check the bore temperature circuit.

TABLE D-1
IDRIVE SAMPLE PROTOCOL

<u>PATIENT REGISTER</u>	[New Pt]	<u>SCANNING RANGE</u>	
<u>PATIENT INFORMATION</u>		FOV.	[32]
Patient Id	geservice	Slice Thickness.	[7]
Patient Name	test	Spacing	[1.5]
Weight (Lb)	111	Start	0
	[Landmark]	End..	0
Landmark	[Sternal Notch]	# Slices	1 (default)
		L/R Center	0 (default)
<u>PATIENT PROTOCOLS</u>	[Patient Position]	P/A Center	0 (default)
		Table Delta	0.00 (default)
<u>PATIENT POSITION</u>		<u>ACQUISITION TIMING</u>	
Patient Position	[>] [Supine]	Freq	[128]
Patient Entry.	[>] [Head First]	Phase	[128]
Coil	[...] [Body] [Accept]	NEX	[1]
		Phase FOV	1.00 (default)
<u>IMAGING PARAMETERS</u>		Freq Dir	[>] [R/L]
Plane	[>] [Axial]	Auto Center Freq	[>] [water]
Mode	[>] [2D]	Autoshim	[on]
Pulse Seq	[...] [Fast GRE] [Accept]	(lowest window)	[Save Series]
Imaging Options	[Real-time]		
Psd Name	no entry		
Protocol	no entry		
<u>SCAN TIMING</u>			
# of Echoes	1 (default)		
TE	[Min Full]		
TR.	[Minimum]		
Flip Angle	[20]		
Bandwidth	[62.50]		

D-2 Signa Power up Delay

When Signa is first powered on there is a considerable period of time that nothing is seen on the monitor; up to 4 minutes depending on the computer type. The system is checking configurations and starting software processes at this time. If you are concerned that the system is hung, push the <Esc> button on the top left side of the keyboard. The system will then post to the monitor all its boot up activities.

D-3 "Configup" Script Modifications To Files

When RestoreINFO is run after upgrading from Release 8.2.5 software the first time, new version configuration files are overwritten. The "configup" script updates the MR System configuration file with the new changes as needed:

1. The "configup" script updates the MRconfig.cfg file with the new "fastprescan" parameter if it is not present (if "fastprescan" parameter is present, it does not add or change it):

```
# This field is added by shivar@wge for MRIge47785
# A fastprescan field to turn Faster Prescan on & off .
#           type = integer, legal values = 0 (off)
#                                           = 1 (on)
fastprescan = "1"
```

2. The "configup" script updates the MRconfig.cfg file with the new "cerdType" parameter if it is not present (if "cerdType" parameter is present, it does not add or change it):

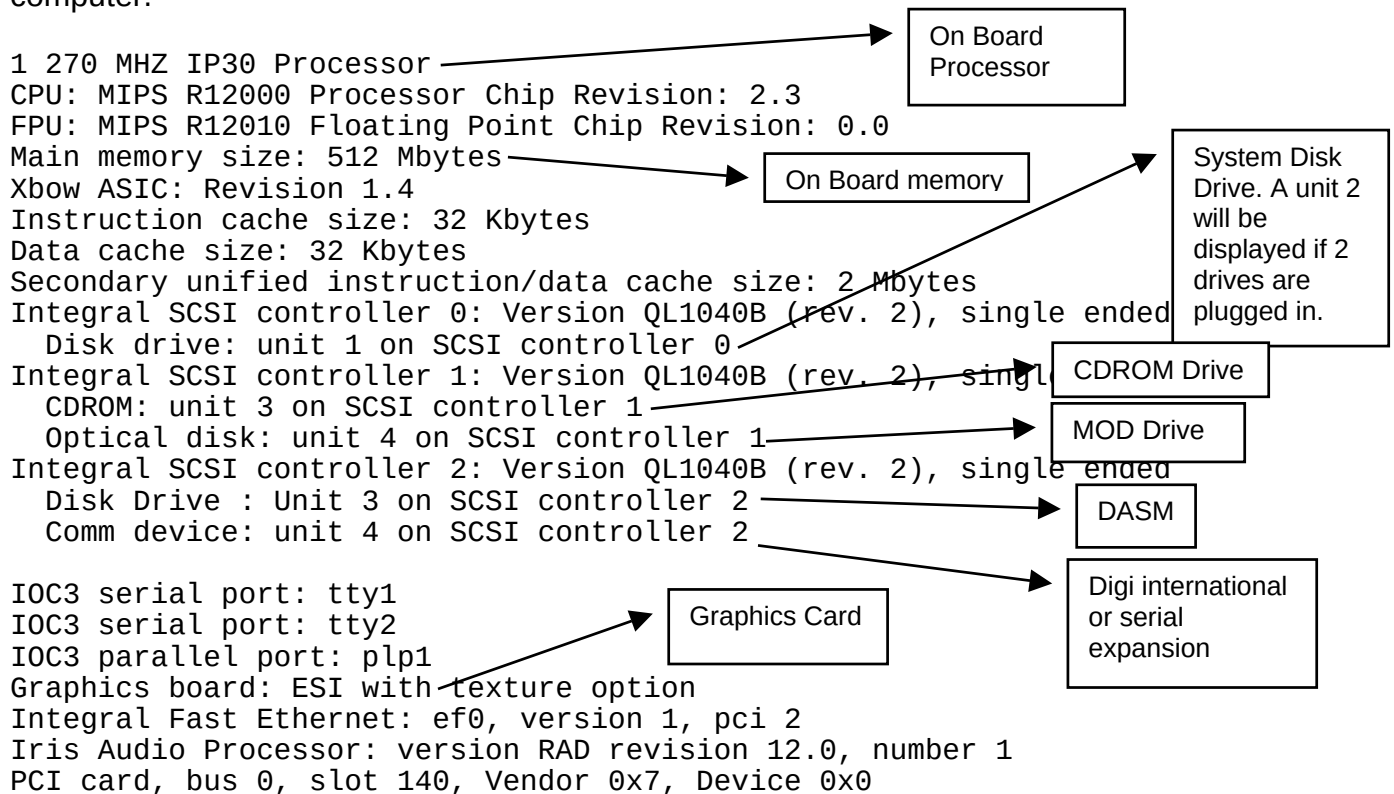
```
# Type of (Universal) Combined Exciter Receiver
# Data Acquisition Board (CERD)
# type = integer
# Usage
# X Y Y
# \ \ \----- (U) CERD Slot #1 (VME slot 13 , right next to
# power supply)
# \ \----- (U) CERD Slot #2 (VME slot 10)
# \----- MKE or YMS Receiver mapping
# Where X =
#   0: MKE receiver mapping scheme
#   1: YMS receiver mapping scheme
#
# Where Y =
#
#   0 : No(U) CERD Installed
#   1 : Narrowband CERD
#   2 : Universal CERD (UCERD)
# Legal Combinations
#
#   0yy: MKE default receiver mapping
#   1yy: YMS default receiver mapping
#   x01: Narrowband CERD on Slot #1
#   x02: Universal CERD on Slot #1
#   x21: Narrowband CERD on Slot #1 ;Universal CERD on Slot #2
#   x22: Universal CERD in both Slot #1 and Slot #2
cerdType = "001"
```


3. The "configup" script updates the GradientConfig.cfg file with the new Full Power SGD parameters if not present (if the following parameters are present, it does not add or change them):

```
# Added by shivar@wge for MRIGe49945
Sdlim = "0.60"
Lcoil = "1.40"
Rac = "152"
Pmgs = "3000"
mGRAMpower = "9000"
```

D- 4 [hinv] Command

The [hinv] command can be used to check the configuration of the operator workspace. To use the command, open a command window and at the prompt type: **hinv <Enter>**. The following is an example of an [hinv] output of a common configured Signa system using Octane computer.



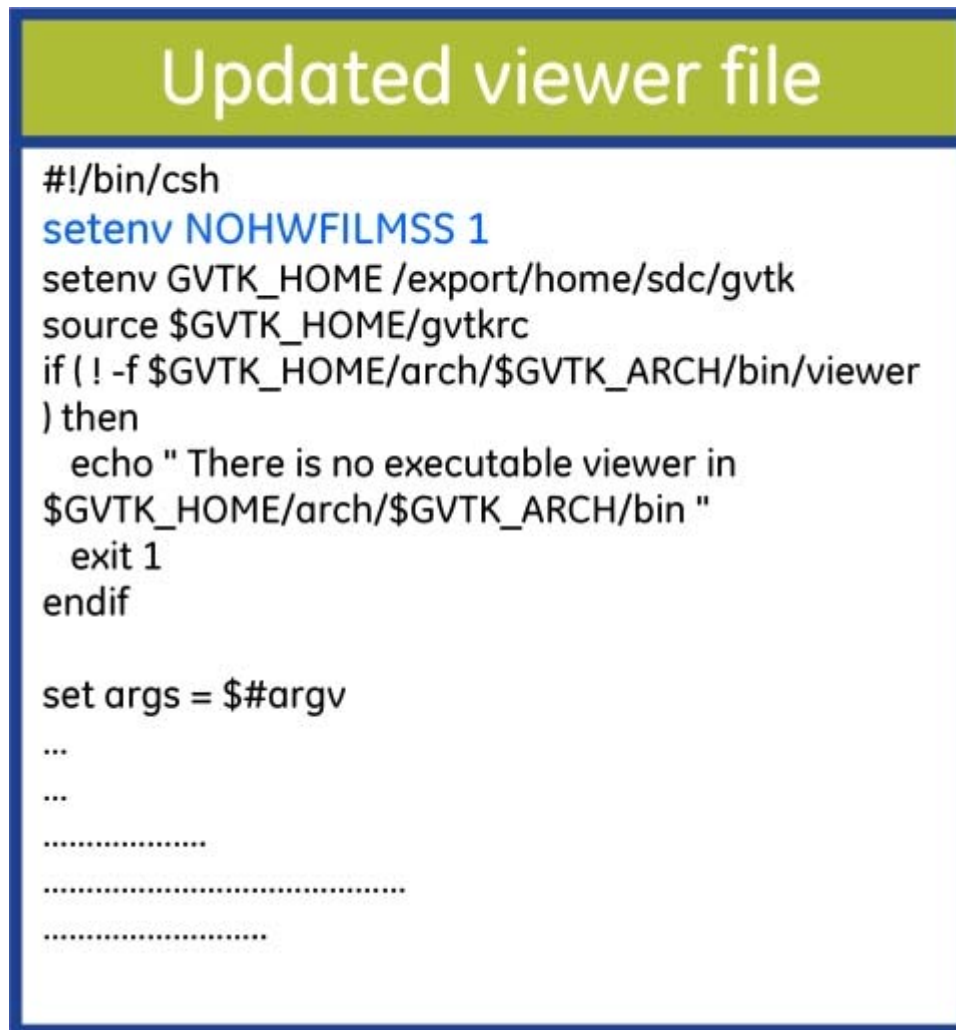
D-5 Film Artifacts Due to HW Issues

Problem Symptom: Film frame contains artifact(s) when filming with film composer using 6 on 1 format and split screen. The artifact will not display on the viewer, but displays on films for a combination of some printers and some video cards.

An environment variable has to be defined in `~sdc/bin/viewerfile`, which will address this issue.

Follow these steps:

1. Quit all Viewers and Miniviewers.
2. Open C-shell.
3. Type: **cd /sdcb/bin.**
4. Type: **cp viewer viewer.lfc.**
5. Type: **jot viewer.**
6. In the first line of this file, enter the following: **setenv NOHWFILMSS 1.**
7. Click on **[File]** on the jot window, select **[Save]**, then exit the jot window.
8. Type: **more viewer** and verify that the file is updated as noted in Illustration D-1.



```
#!/bin/csh
setenv NOHWFILMSS 1
setenv GVTK_HOME /export/home/sdc/gvtk
source $GVTK_HOME/gvtrc
if ( ! -f $GVTK_HOME/arch/$GVTK_ARCH/bin/viewer
) then
    echo " There is no executable viewer in
$GVTK_HOME/arch/$GVTK_ARCH/bin "
    exit 1
endif

set args = $#argv
...
...
.....
.....
.....
```

UPDATED VIEWER FILE
ILLUSTRATION D-1

APPENDIX E - SERVICE SOFTWARE LOAD

E-1 Loading Service Software

1. Go to the install GUI window, and select the **ServiceSW/Picture this/InSite** tab.
2. Proceed to Section E-2, Load Advanced Service Software, **or** Section E-3, Load Restricted Service Software (for sites with GE Service Only).

(Restricted Service Software is only available for GE use; Advanced Service Software is only available to sites with a valid Advanced Service Package Limited License.)

E-2 Load Advanced Service Software

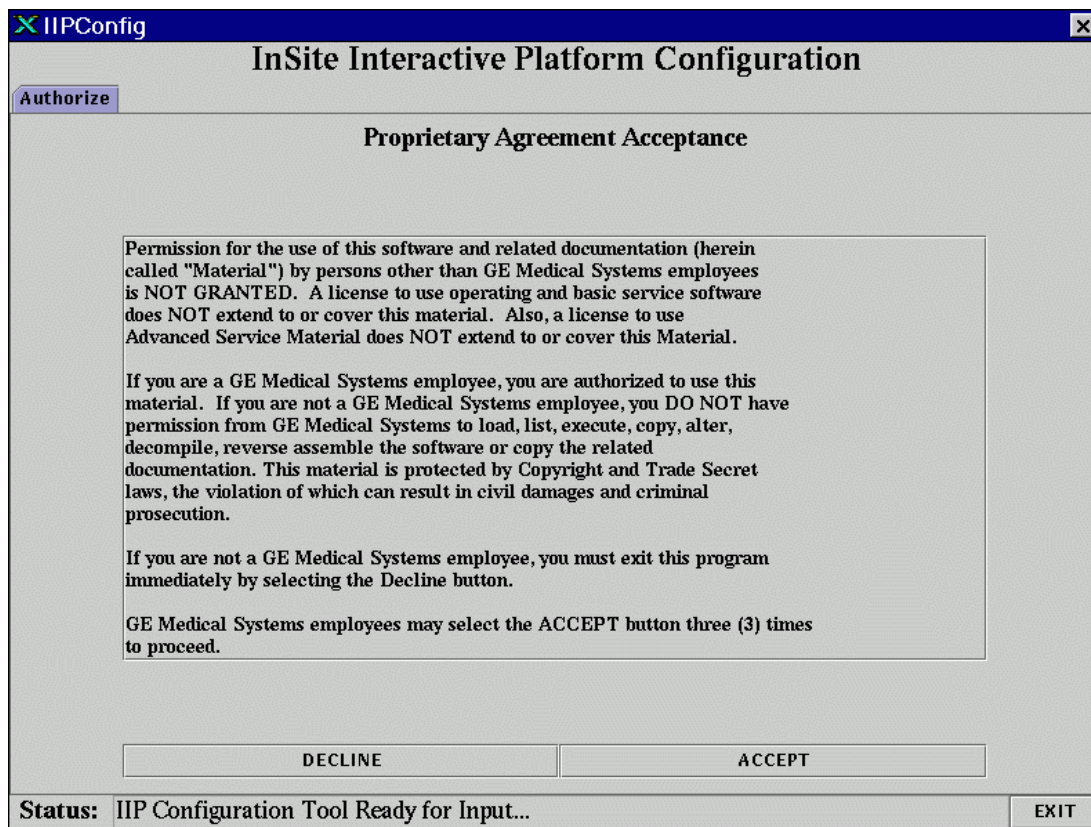
To load Advanced Service Software on the system:

1. Place the Advanced Service CD-ROM 2167558-16, Rev. 9.1, in the host CD-ROM drive.
2. Select **[Load Advanced Service]**.
3. Read the proprietary message window, and respond accordingly.
4. A message "Advance software load in progress..." will display. The load will take about 5 minutes.
4. Wait for the message "Advanced Service load done" in the Install GUI status bar (at bottom of window).
5. Proceed to Section 6-5, Option Keys and Verification of Restored Information.

E-3 Load Restricted Service (for sites with GE Service only)

To load Restricted Service Software on the system:

1. Place Restricted Service CD-ROM 2167557-29, Rev 9.1, in the host CD-ROM drive.
2. Select **[Load Restricted Service]**. Read the proprietary message and respond accordingly.
3. A message "Restricted software load in progress..." will display. The load will take about 14 minutes. Be patient. Illustration E-1 will display near the end of the Restricted software load.



PROPRIETARY AUTHORIZATION TAB

ILLUSTRATION E-1

4. GEMS Field Engineers loading GEMS Restricted Service Software should configure InSite at this time. Select **[Accept]** 3 times and proceed to Section E-4, Configuration for Prodiags and InSite Checkout.

If you do not have an InSite modem or you will configure it later, select **[Exit]**, and proceed to Section E-5, Option Keys and Verification of Restored Information.

E-4 Configuration for ProDiags and InSite Checkout (for sites with GE Service)

The instructions in this section will step you through the minimum amount of screens to enable the basic functionality needed to enable ProDiags and complete an InSite Checkout with the OnLine Center. For information on the CUSTOM setup in the InSite Configuration GUI, refer to the procedure "Insite/IIP Installation Signa 8.3 or above" on the Signa 8.x Service Methods CD-ROM (2160623-1).

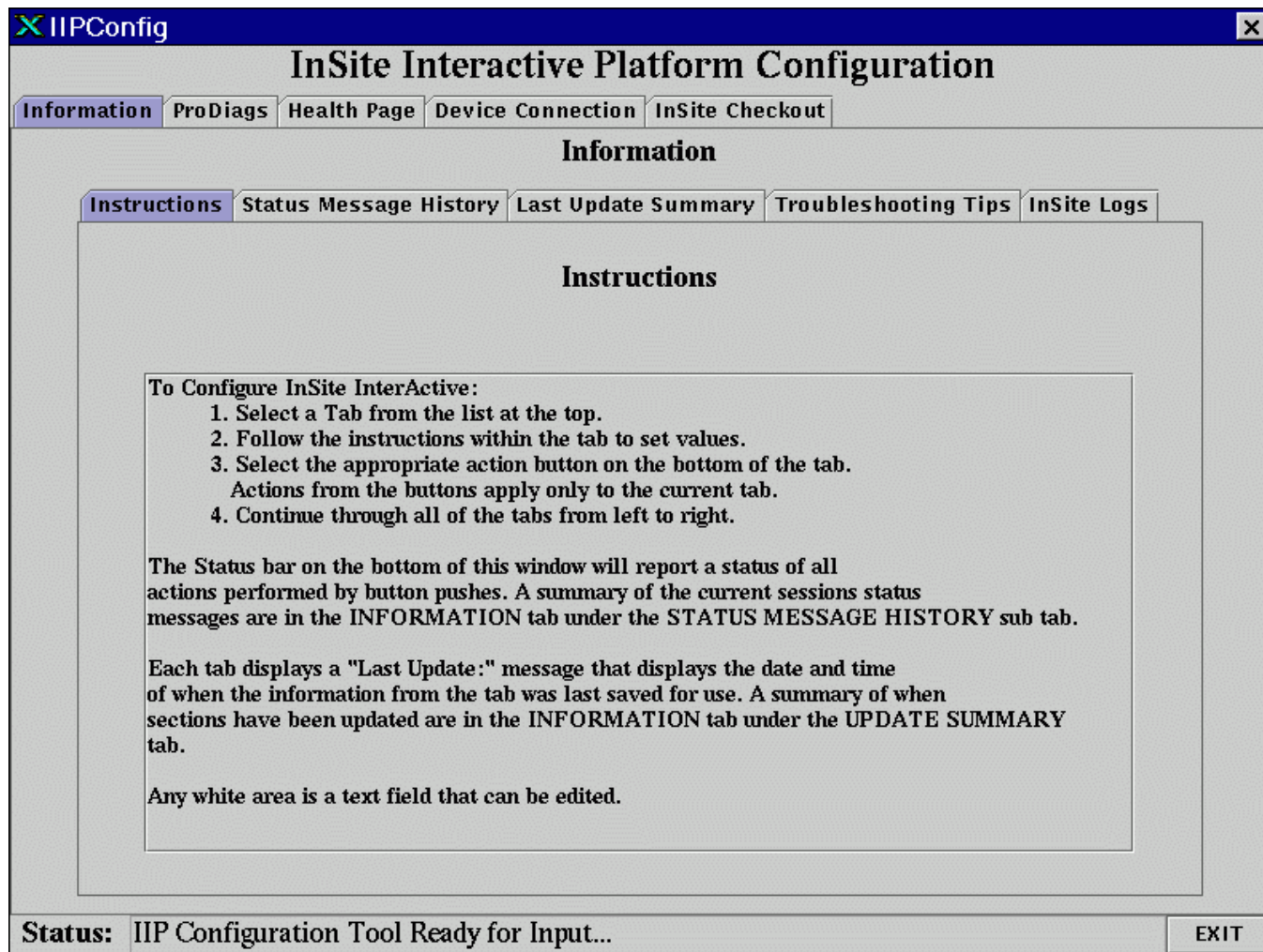
Note

Pro Diags are defaulted to call out to the ASC daily. If the attempt to call out fails, the software re-attempts to call out hourly until successful. If the modem or network connection is down, **these hourly attempts to call out can slow down the scanner**. It is suggested that during InSite IIP setup, ProDiags call out to ASC be configured to dial out weekly and reattempt daily.

Note

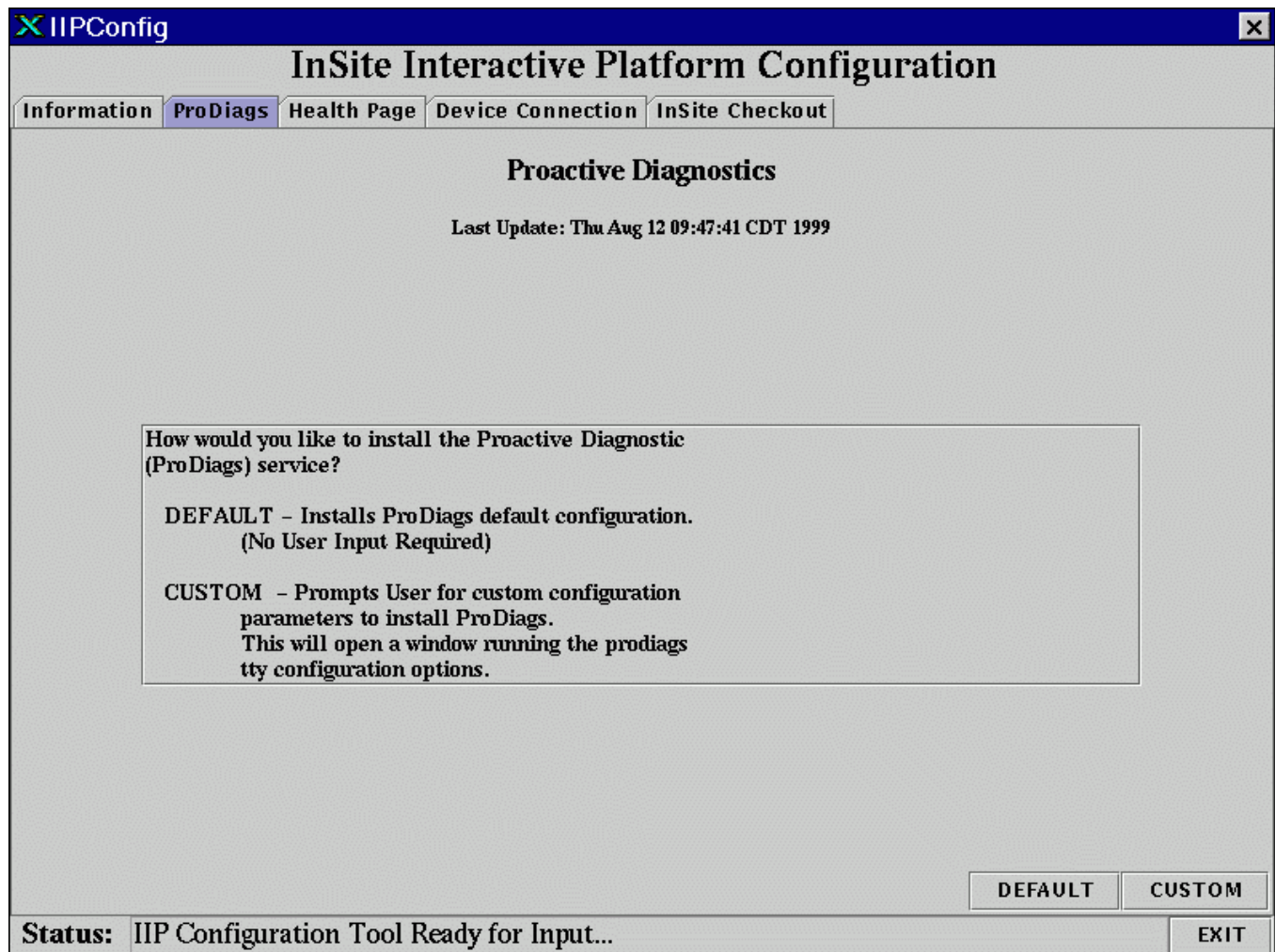
Due to software changes at the OLC to support iLinq (formerly called InSite Interactive), all 9.x systems must initially do a Manual Checkout (i.e., **FE must call the OLC for initial checkout.**) Once this is done, the Auto Checkout feature can be used for future reloads.

1. Following acceptance, the main tabs are initialized. See Illustration E-2.



INFORMATION TAB, INSTRUCTIONS SUB-TAB
ILLUSTRATION E-2

2. Select **ProDiags** tab to display screen as shown in Illustration E-3.
3. Select **[Default]**. This will configure the ProDiags feature with default schedules provided with the software. The Health Page screen will then be displayed. See Illustration 6-4. A status message "ProDiags Auto Setup: Completed Successfully" should be displayed at the bottom of the screen.



PROACTIVE DIAGNOSTICS TAB
ILLUSTRATION E-3

The screenshot shows a window titled "IIPConfig" with a subtitle "InSite Interactive Platform Configuration". It has five tabs: "Information", "ProDiags", "Health Page" (which is selected), "Device Connection", and "InSite Checkout". The "Health Page" tab contains the following text: "Health Page", "Last Update: Wed Aug 18 09:56:36 CDT 1999", a text box with instructions to enter or edit email addresses, a note about including the full internet address, an example email address, and two options: "DEFAULT" and "CUSTOM". Below this is a checkbox labeled "HealthPage Report Disabled" and a large text box labeled "Enter Address List". At the bottom right are buttons for "DEFAULT" and "CUSTOM >>". A status bar at the bottom left says "Status: IIP Configuration Tool Ready for Input..." and a button labeled "EXIT" is on the bottom right.

IIPConfig

InSite Interactive Platform Configuration

Information ProDiags **Health Page** Device Connection InSite Checkout

Health Page

Last Update: Wed Aug 18 09:56:36 CDT 1999

Enter or edit E-mail addresses to receive the Health Page report.
You MUST enter at least one address.

NOTE: Include the full Internet address or you will not receive the report!
Example: FirstName.LastName@med.ge.com

DEFAULT - Updates the Health Page E-mail addresses only.

CUSTOM - To gain access to Custom Health Page settings.
NOTE: E-mail addresses MUST be entered before
entering Custom Health Page

☐ **HealthPage Report Disabled**

Enter Address List

DEFAULT CUSTOM >>

Status: IIP Configuration Tool Ready for Input... EXIT

HEALTH PAGE TAB
ILLUSTRATION E-4

- Click in the **Enter Address List** box and type in a valid e-mail address. A valid e-mail address is defined as a string in the format **<name> @ <location>**

The format for GEMS e-mail addresses is: *<firstname> . <lastname> @med.ge.com*

Be careful to enter correct e-mail address(es). All e-mail addresses are saved to the */usr/g/config/healthpg.adr* file in the format: *user<#>=<email address>*.

- Select the **[DEFAULT]** button to configure Health Page with the default schedule file. The Device Connection screen will then be displayed. See Illustration E-5. A status message "Health Page Email... Completed Successfully" should be displayed.

InSite Interactive Platform Configuration

Information ProDiags Health Page **Device Connection** InSite Checkout

Connection Device Configuration

Last Update: NOT Configured

Device Connection Type Modem ▼

Select the InSite Connection Type above, or select the Modem settings given below and Press

APPLY – To update the modem and modem configuration files with model and country defaults.

CUSTOM – For other modem settings.

Dial-out Prefix
9,1 ▼

Internal Prefix
None ▼

Dialing mode
Tone ▼

Modem Type
MultiTech MT5634ZBA V.90 ▼

Country
Default – All Others ▼

CPU Serial Port Name
/dev/ttyf1 ▼

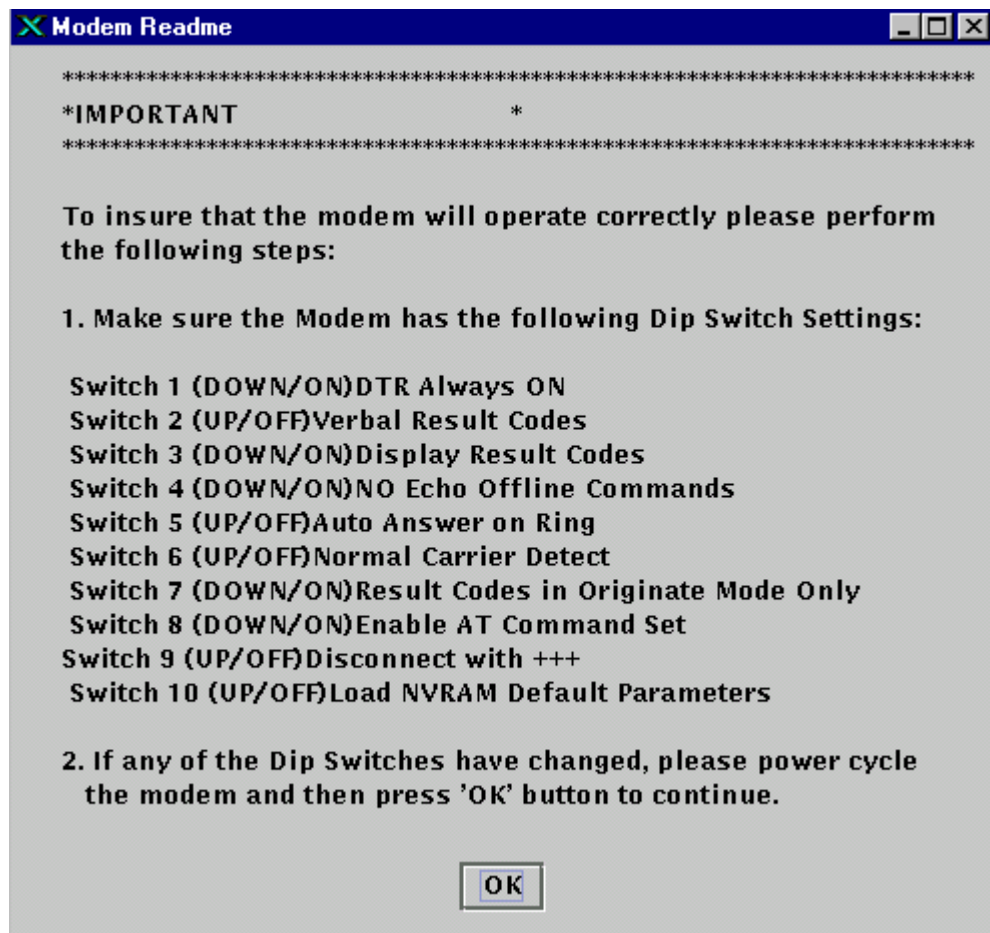
CPU Serial Port Speed
38400 ▼

APPLY **CUSTOM**

Status: IIP Configuration Tool Ready for Input... **EXIT**

DEVICE CONNECTION TAB
ILLUSTRATION E-5

6. Select a Dial-out prefix or type in a site-specific prefix. Dial-out prefixes may be required on some sites to get a line outside of the hospital. " 9,1 " is the most common and is the default.
7. Internal Prefix should not be used (See Dial Out Prefix). "None" must be selected for proper operation.
8. Select Modem Type that you have at this site. Note: Modem Type was recorded in Table 2-1.
9. Select **[APPLY]** to setup registers on the modem, and store values to the modem's NVRAM. When the **[APPLY]** button is selected, another screen will be displayed for the selected modem type (see example in Illustration E-6).

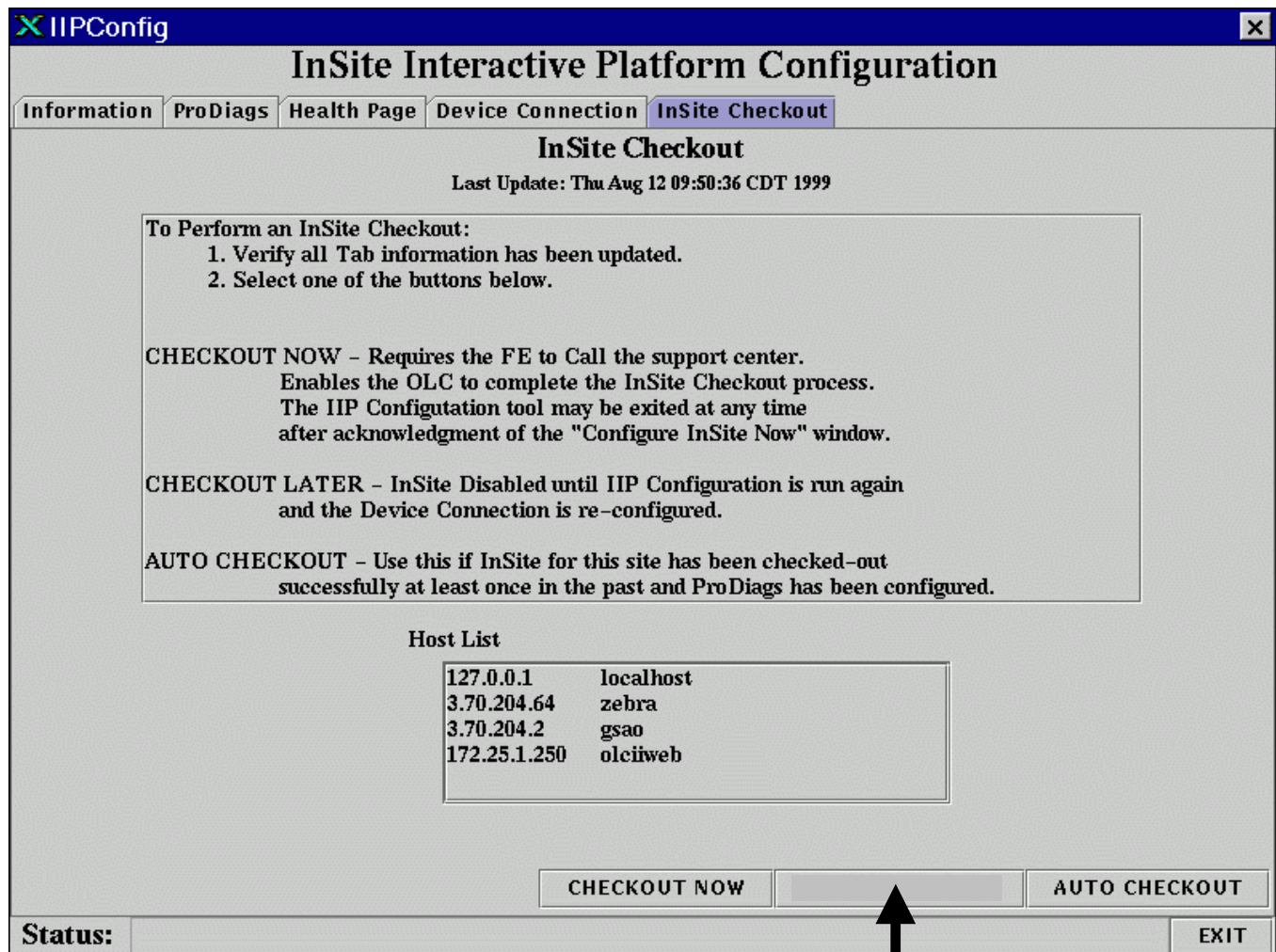


MODEM "README" SCREEN (EXAMPLE IS FOR USR V.34 SELECTION)
ILLUSTRATION E-6

NOTE

Be aware the InSite Configuration may recommend different switch settings for your modem as compared to the settings that worked with previous software. Verify the modem switch settings match those recommended in the InSite Configuration.

9. After properly verifying/configuring any switches or other modem parameters on the modem, select **[OK]** which will do the programming of the modem. The InSite Checkout screen will then be displayed. See Illustration E-7.



INSITE CHECKOUT TAB
ILLUSTRATION E-7

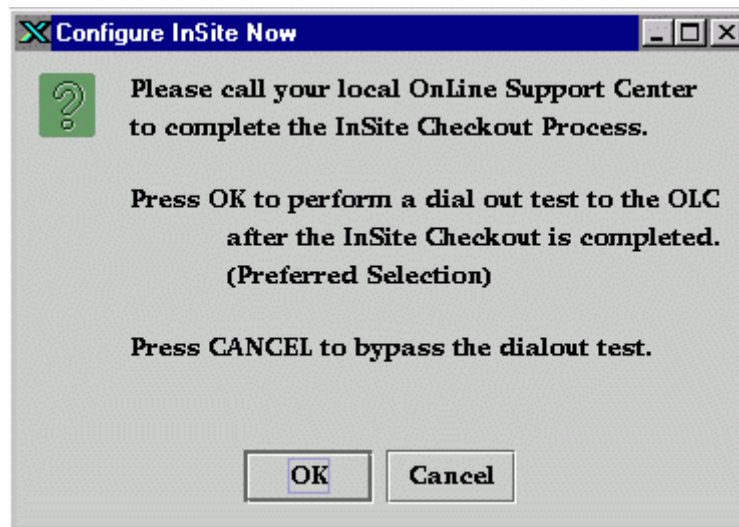
Disable Insite (ASP2 or later SW)

Checkout Later (8.3 Software)

10. The InSite Checkout Tab allows the user to complete the InSite configuration process. The three checkout options are described below, but you must select [CHECKOUT NOW].

- a) **[CHECKOUT NOW]** configures PPP to allow a dial in connection from the OnLine Center. The user must call the local OnLine Center to complete the checkout. See Illustration E-8.

If an InSite Checkout cannot be completed at this time, after selecting **[CHECKOUT NOW]**, select **[Cancel]** followed by **[Exit]**. This will bring up the Exit Confirmation Window (Illustration E-9). Select **[OK]** to exit the InSite Configuration GUI. Exiting in this manner will leave the serial port of the computer enabled allowing your OnLine Center to complete the checkout at a future time without requiring an FE be onsite.



CONFIGURE INSITE NOW REQUESTER
ILLUSTRATION E-8

Note

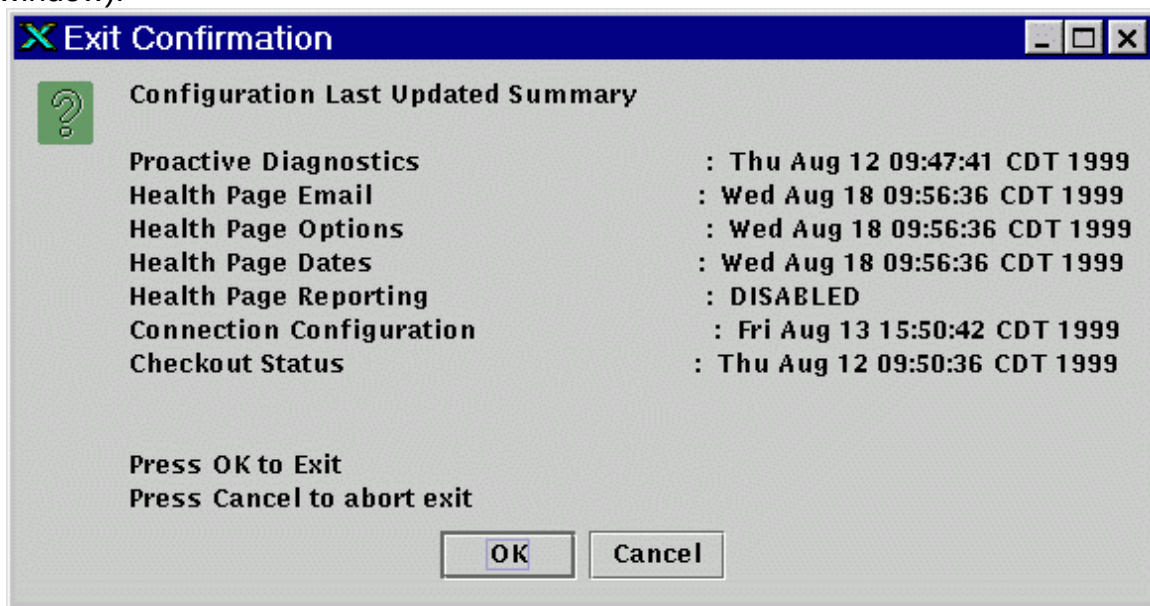
When the user selects the **[CHECKOUT NOW]** button, the “Configure InSite Now” window is displayed. This provides the user the option to run a dial-out test upon completion of the InSite Checkout Process. If the user selects the **[OK]** button before the InSite Checkout Process is complete, the IIP Configuration Tool watches for a `~insite/sclink.cfg` file to be downloaded over the connection. The IIP Configuration Tool will only wait 10 minutes for the `sclink.cfg` file to arrive. When the file displays, the dial-out test starts. This test ensures the dialout prefix provided by the user and the number of the OnLine Center are correct. This also confirms that Proactive Diagnostics messages and **[Contact GE]** requests can be sent to the OnLine Center for sites entitled for Insite Interactive.

When the dialout test is in the process of connecting to the AutoSC, the IIP Configuration GUI will not update and may appear locked up until the connection to the AutoSC is completed or the connection times out. Please be patient.

- b) **[CHECKOUT LATER]** (8.3 and ASP1Software), **[Disable Insite]** (ASP2 Software). **Never select this** because it disables PPP and will require an FE to be on-site to invoke this screen for the OnLine Center to be able to complete an InSite Checkout. If this is selected by mistake, the user **MUST** rerun the IIP Configuration Tool, select the **[APPLY]** button on the Modem Tab, and then select the **[CHECKOUT NOW]** button to enable PPP for dial in.
- c) **[AUTO CHECKOUT]** is not used for this FMI because a manual checkout is required to be updated to the latest model type in the OLC database. The RestoreINFO process does restore the `/usr/g/insite/sclink.cfg` file needed for Auto Checkout to work. This button will only be active if a `sclink.cfg` file exists in the directory above.

[AUTO CHECKOUT] will run ConfigLink, dial the AutoSC, queue a task with the AutoSC to dial back, and wait for the AutoSC to dial back and leave a file `/tmp/autocheckout`. The IIP Configuration Tool will wait only 10 minutes for the file to display. Typically, the AutoSC will call back in 5-6 minutes.

11. After selecting **[Exit]**, an updated summary is provided. See Illustration E-9. A valid date/time should be displayed for each function that you successfully configured. If any function shows **"NO FILE"**, that function was not properly configured.
12. Select the **[OK]** button to exit IIP Configuration Tool.
13. Wait for the message "Restricted Service load done" in the Install GUI status bar (at bottom of window).



EXIT CONFIRMATION REQUESTER
ILLUSTRATION E-9

Important: For additional information about setting up or troubleshooting the Insite software/hardware, refer to the procedure "Insite/IIP Installation Signa 8.3 or above" on the Signa 8.x Service Methods CD-ROM (2160623-1).

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASON FOR CHANGE
A	Oct, 2002	Kargard	M3 release for 9.1 software.
0	April, 2003	Kargard	Initial release for 9.1 software.
1	Dec, 2003	Stetz	IIP Config Changes.
2	March, 2005	Kargard	Minor changes incorporated for CheckINFO script & others.
3	May, 2008	Morrell	Section D-5 "Film Artifacts due to HW Issues" added